# Game Theory (Core)

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# 1 Static Games with Complete Information

## 1.1 Representation of a Game

**Definition 1.1 (Strategic Game).** A strategic game consists of

- a finite set $N$ (the set of players)
- for each player $i \in N$ a nonempty set $A_i$ (the set of actions available to player $i$)
- for each player $i \in N$ a preference relation $\succsim_i$ on $A = \times_{j \in N} A_j$ (the preference relation of player $i$) and thus individual payoffs associated with $u_i(a_i, a_{-i}), a_i \in A_i, a_{-i} \in \Pi_{j \neq i} A_j$

**Remark.** If the set $A_i$ of actions of every player $i$ is finite then the game is finite

**Matrix Form Representation**
Here are some famous examples of games represented in matrix form

1. Prisoner's Dilemma

| | $d$ | $c$ |
|---|---|---|
| $D$ | $(1,1)$ | $(3,0)$ |
| $C$ | $(0,3)$ | $(2,2)$ |

2. Battle of the Sexes

| | $b$ | $s$ |
|---|---|---|
| $B$ | $(2,1)$ | $(0,0)$ |
| $S$ | $(0,0)$ | $(1,3)$ |

3. Matching Pennies

| | $d$ | $c$ |
|---|---|---|
| $D$ | $(1,-1)$ | $(-1,1)$ |
| $C$ | $(-1,1)$ | $(1,-1)$ |

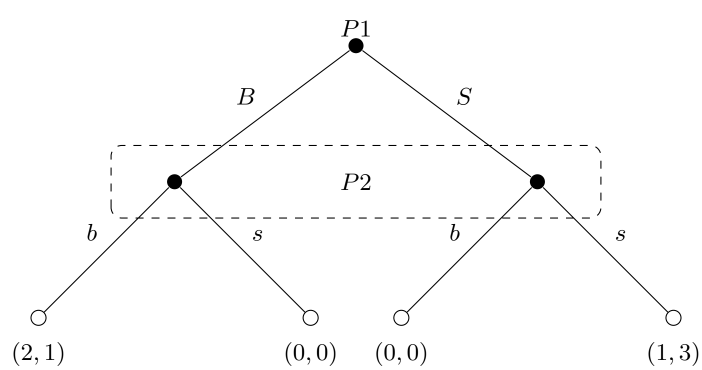
**Remark.** The row player's payoff is the first entry in the payoff vector while the column player's payoff is the second entry in the payoff vector

**Extensive form representation**
We may represent a simultaneous game as an extensive form game.

**Example 1.2.** Consider the Battle of the Sexes game

| | $b$ | $s$ |
|---|---|---|
| $B$ | $(2,1)$ | $(0,0)$ |
| $S$ | $(0,0)$ | $(1,3)$ |

Then, we can represent it as an extensive form game like so




**Definition 1.3 (Strategy).** A strategy is a complete (contingent) plan, which specifies how the player will act in every possible distinguishable circumstance in which she might be called upon to move.

**Definition 1.4 (Pure Strategies in a Static Game of Complete Information).** Let $G = \langle N, (A_i), (\succsim_i)\rangle$ be a strategic game. We denote a member $a_i \in (A_i)$ as a pure strategy of player $i$

*Technical Note*: Initially, the preference relation $\succsim_i$ of each player $i$ is defined over the set $A_i = \times_{i \in N} A_i$ of action profiles. To account for the ability to randomise actions, we assume that the preference relation of each player $i$ satisfies the assumptions of Von Neumann and Morgenstern, so that it can be represented by the expected value of some function $u_i : A \to \mathbb{R}$. Then, we represent our strategic game as the triple $G = \langle N, (A_i), (u_i)\rangle$ with $u_i : A \to \mathbb{R}$ for each $i \in N$ as a function whose expected value represents player $i$'s preferences over the set of lotteries on $A$.

**Definition 1.5 (Mixed Strategies in a Static Game of Complete Information).** Let $G = \langle N, (A_i), (u_i)\rangle$ be such a strategic game. We denote by $\Delta(A_i)$ the set of probability distributions over $A_i$ and refer a member $a_i \in \Delta(A_i)$ as a mixed strategy of player $i$

## 1.2 Rationalizability, Dominance, Iterated Elimination of Strictly Dominated Strategies

**Definition 1.6 (Strictly Dominated Strategies).** A (pure) strategy $\alpha_i \in A_i$ of player $i$ is strictly dominated iff there is a mixed strategy $\beta_i$ such that for all (pure) action profiles $a_{-i} \in A_{-i}$

$$u_i(\beta_i, a_{-i}) > u_i(\alpha_i, a_{-i})$$

**Remark.** Notice that we only consider pure strategies being strictly dominated; furthermore, an action can easily be strictly dominated without being strictly dominated by a pure action.

**Example 1.7 (Dominated by Mixed Strategy).** Consider the following normal form game

| | $L$ | $M$ | $R$ |
|---|---|---|---|
| $U$ | $(0,0)$ | $(1,1)$ | $(0,3)$ |
| $D$ | $(1,3)$ | $(0,1)$ | $(1,0)$ |

No pure action strictly dominates M: M is better than L against U, and better than R against D. But M is strictly dominated by a mixed action, namely the 50–50 mix of L and R: this mixed action yields an expected payoff of 1.5 regardless of player 1's choice, strictly better than action M's payoff of 1

**Proposition 1.8 (Pearce's Lemma).** An action of a player is never-best (for any of belief over opponents' strategy) if and only if it is strictly dominated.

The intuition here is that undominated strategies are always sometimes-best (in the weak sense). We can verify in **Example 1.7** that M is never-best and strictly dominated while L and R are sometimes best and not strictly dominated.

**Remark.** The "if" part of the claim holds generally, including for non-finite (action space i.e. $|A_i| \geq \aleph_0$ for any $i \in \mathcal{I}$) games while the "only if" part holds only under the assumption of finite games and does not hold generally for non-finite games. In other words, in non-finite games, it is not the case that if is never best, it is strictly dominated.

**Definition 1.9 (Strictly Dominant Strategies).** For mixed actions $\alpha_i, \beta_i \in \Delta(A_i)$ of a player $i$, we say that $\alpha_i$ strictly dominates $\beta_i$ iff $u_i(\alpha_i, a_{-i}) > u_i(\beta_i, a_{-i})$ for every (pure) profile $a_{-i} \in A_{-i}$ of other players' actions.

We have seen that the (only) implication of rationality and comprehension of the game is that players will not choose never-best (equivalently, strictly dominated) actions. But what if each player additionally knows that her opponents are rational? If each player knows that every player is rational and also know that other players know that every player is rational, then we say that there is *common knowledge of rationality.* The implication of *common knowledge of rationality* is captured by rationalisability.

**Definition 1.10 (Rationalisability).** For each player $i \in I$ and each conjecture $\alpha_{-i} \in \Delta(A_{-i})$, write $\mathrm{BR}_i(\alpha_{-i}) \subseteq A_i$ for the set of best responses to $\alpha_{-i}$. Define $X_i^0 := A_i$ and $X_{-i}^0 := A_{-i}$ for each $i \in I$. For $t \in \mathbb{N}$, iteratively define

$$X_i^t := \mathrm{BR}_i(\Delta(X_i^{t-1})) \equiv \bigcup_{\alpha_{-i} \in \Delta(X_{-i}^{t-1})} \mathrm{BR}_i \alpha_{-i}, \quad \text{for each } i \in I$$

$X_{-j}^t := \Pi_{i \in I \setminus \{j\}} X_i^t$ for each $i \in I$, and $X^t := \Pi_{i \in I} X_i^t$. Finally define $X_i^\infty := \bigcap_{t \in \mathbb{R}} X_i^t$ for each $i \in I$ and $X^\infty := \Pi_{i \in I} X_i^\infty$. An action $\alpha_i \in A_i$ of player $i$ is called rationalisable iff it belongs to $X_i^\infty$, similarly an action profile $a = (a_i)_{i \in I} \in A$ is called rationalisable iff it belongs to $X^\infty$

Another way to phrase this is that a pure strategy of player $i$ is rationalisable if it is a best response against some mix/belief of the opponents' rationalisable strategies. We may express rationalisability in the following (more tractable) way.

**Proposition 1.11.** The action $a_i^*$ of player $i$ is rationalisable if for each player $j$ there exists a set $Z_j \subseteq A_j$ such that

- $a^* \in Z_i$
- for every player $j$, every action $a_j^*$ maximises the expected payoff of player $j$ given their belief $p_j$ that assigns a positive probability only to action profiles in $Z_{-j}$

**Remark.** I like **Definition 1.10** because it shows the connection between *iterated deletion of never-best actions* (or strictly dominated actions by Pearce's Lemma) because IESDS is precisely the procedure laid out in the definition.

**Proposition 1.12.** In finite games the set of correlatedly rationalizable actions coincides with the set of actions surviving the iterated elimination of strictly dominated strategies.

**Corollary 1.13.** In two player finite games, the set of rationalizable actions coincides with the set of actions surviving iterated elimination of strictly dominated strategies

*Proof.* Directly follows because no need to think about correlations in two player games. ■

**Example 1.14.** Consider a three-player game where player 3 chooses among $W, X, Y, Z$, and players 1 and 2 choose rows ($U/D$) and columns ($L/R$) respectively. The payoff matrices (to all three players) for each of player 3's strategies are:

| $W$ | $L$ | $R$ |
|---|---|---|
| $U$ | $8,8,8$ | $0,0,0$ |
| $D$ | $0,0,0$ | $0,0,0$ |

| $X$ | $L$ | $R$ |
|---|---|---|
| $U$ | $4,4,4$ | $0,0,0$ |
| $D$ | $0,0,0$ | $4,4,4$ |

| $Y$ | $L$ | $R$ |
|---|---|---|
| $U$ | $0,0,0$ | $0,0,0$ |
| $D$ | $0,0,0$ | $8,8,8$ |

| $Z$ | $L$ | $R$ |
|---|---|---|
| $U$ | $3,3,3$ | $3,3,3$ |
| $D$ | $3,3,3$ | $3,3,3$ |

**$X$ is not rationalizable under independent (product) beliefs.**
Let $\Pr(U) = p$ and $\Pr(L) = q$. Under product beliefs, player 3's expected payoff from $X$ is:

$$v_3(X) = 4pq + 4(1-p)(1-q).$$

For $X$ to be a best response, it must not be dominated by $W$ or $Y$, which requires:

$$4pq + 4(1-p)(1-q) \geq \max\{8pq,\, 8(1-p)(1-q)\} \implies p = q = \tfrac{1}{2}.$$

However, at $p = q = 1/2$ we get $v_3(X) = 2 < 3 = v_3(Z)$, so $X$ is dominated by the safe strategy $Z$ and is therefore not rationalizable under product beliefs.

**$X$ is rationalizable under correlated beliefs.**
Now suppose player 3 holds correlated beliefs over the strategies of players 1 and 2. Let $\Pr(U, L) = p'$ and $\Pr(D, R) = q'$, with the remaining probability $1 - p' - q'$ placed on $(U, R)$ and $(D, L)$ (which yield a payoff of 0 under $X$). Player 3's expected payoff from $X$ is then:

$$v_3(X) = 4p' + 4q'.$$

For $X$ to not be dominated by $W$ or $Y$, we need:

$$4p' + 4q' \geq \max\{8p',\, 8q'\} \implies p' = q' = \tfrac{1}{2}.$$

At $p' = q' = 1/2$ we obtain $v_3(X) = 4 > 3 = v_3(Z)$, so $X$ is not dominated by $Z$ either. Hence, $X$ *is* rationalizable once we allow for correlated beliefs.
This example illustrates that the set of rationalizable strategies can be strictly larger under correlated beliefs than under the independence restriction imposed by product beliefs.

**Definition 1.15 (Iterated Deletion of Strictly Dominated Strategies).** We define the iterated deletion of strictly dominated strategies as follows:

1. Let $\Sigma_i(R)$ be a set of mixed strategies with support $R$ e.g. $\Sigma(\{A, C\}) = \{pA + (1-p)C\}$
2. For all players $i$ define $R_i^0 = A_i$ i.e. the set of pure strategies

3. Let $R_i^k \subseteq R_i^{k-1}$ be a subset of undominated actions, i.e. such that for any $a_i \in R_i^k$ there is no such $\alpha_i \in \Sigma(R_i^{k-1})$ that $u_i(\alpha_i, a_{i-}) > u_i(a_i, a_{-i})$ for all $a_{-i} \in R_{-i}^{k-1}$
4. That is, $R_i^k$ is a set of pure strategies which are not strictly dominated by another (possibly mixed) strategy from $R_i^{k-1}$
5. The set of pure strategies which survive the iterated elimination of strictly dominated strategies is denoted
$$R_i^\infty = \bigcap_k R_i^k$$

**Definition 1.16 (Dominance Solvable Games).** If $R_i^\infty$ is a singleton, then the game is called dominance solvable

**Proposition 1.17.** The order of elimination in IESDS does not matter

**Example 1.18 (IESDS and Continuous Actions).** $N$ players simultaneously choose a number $x_i \in [0, 100]$. They players wish to maximise monetary payoffs. The player whose choice is closest to a proportion $\rho \in (0, 1)$ of the average of the $N$ chosen numbers wins 10 pounds. In the case of ties, the prize is shared equally. The players only choose pure strategies and are not allowed to randomise. Show that all players choosing zero is the only strategy profile that survives iterated deletion of weakly dominated strategies.
*Solution*: Consider $R^0 = [0, 100]$. Hence, $\bar{x} \in [0, 100]$, and $\rho\bar{x} \in [0, 100\rho]$. Player $i$ wants to minimise $|x_i - \rho\bar{x}|$. Since $|x_i - \rho\bar{x}| \geq |100\rho\bar{x}|$ for any $x_i > 100\rho$, so $x_i = 100\rho$ weakly dominates $x_i > 100\rho$ (weakly because $x_i = 100\rho$ may not win either so same payoff at 0). So, $R^1 = [0, 100\rho], \bar{x} \in [0, 100\rho]$ and $\rho\bar{x} \in [0, 100\rho^2]$. So choosing $x_i > 100\rho^2$ is weakly dominated. After $k - 1$ rounds,

$$R^{k-1} = [0, 100\rho^{k-1}]$$

and $x_i > 100\rho^k$ is weakly dominated. So $R^k = [0, 100\rho^k]$. Since $\rho \in (0, 1)$, $\lim\limits_{k\to\infty} \rho^k = 0$ hence

$$\bigcap_{k\geq 0}^{\infty} R^k = R^\infty = \{0\}$$

**Definition 1.19 (Weakly Dominated Strategies).** Strategy $a_i$ is called weakly dominated if there is another strategy $\alpha_i'$ such that for any $a_{-i}$

$$v_i(\alpha_i', a_{-i}) \geq v_i(a_i, a_{-i})$$

moreover, there is $a_{-i}'$ such that

$$v_i(\alpha_i', a_{-i}') > v_i(a_i, a_{-i}')$$

## 1.3 Nash Equilibrium

**Definition 1.20 (Pure NE).** A Nash Equilibrium in pure strategies is an action profile $a^* = \{a_1^*, \ldots, a_n^*\}$ such that for any action $a_i$ of any player $i$

$$u_i(a^*) \geq u_i(a_i, a_{-i}^*)$$

i.e. no unilateral profitable deviation

**Definition 1.21 (Best Response).** The best response correspondence $B_i$ of a player $i$ is

$$B_i(a_{-i}) = \{a_i \in A_i : u(a_i, a_{-i}) \geq u_i(a_i', a_{-i}), \forall a_i' \in A_i$$

**Proposition 1.22.** The set of Nash Equilibrium coincides with the intersection of Best Response correspondences. Formally, the action profile $a_i^*$ is a Nash equilibrium if and only if

$$a_i^* \in B_i(a_{-i}^*), \forall i \in N$$

*Proof.* We prove this bidirectionally. Suppose $\forall i : a_i^* \in B_i(a_{-i}^*)$. Then for any player $i$

$$u_i(a_i^*, a_{-i}^*) \geq u_i(a_i, a_{-i}^*)$$

thus $a^*$ is NE by definition. Suppose $a^*$ is an NE but for some player $i$, $a_i^* \notin B_i(a_{-i}^*)$. Then there exists such $a_i'$ that $u_i(a_i', a_{-i}^*) > u_i(a_i^*, a_{-i}^*)$. Therefore, $a^*$ is not NE by definition. ∎

**Example 1.23 (Games with Finite Actions).**

**Game 1:**

| | $L$ | $C$ | $R$ |
|---|---|---|---|
| $T$ | $0,2$ | $2,0$ | $0,0$ |
| $M$ | $2,0$ | $0,2$ | $0,0$ |
| $B$ | $0,0$ | $0,0$ | $1,1$ |

**Game 2:**

| | $L$ | $C$ | $R$ |
|---|---|---|---|
| $T$ | $1,1$ | $1,1$ | $0,0$ |
| $B$ | $0,0$ | $0,0$ | $1,1$ |

**Game 3:**

| | $L$ | $R$ |
|---|---|---|
| $T$ | $1,2,3$ | $0,0,0$ |
| $B$ | $0,0,0$ | $0,0,0$ |

G1

| | $L$ | $R$ |
|---|---|---|
| $T$ | $0,0,0$ | $0,0,0$ |
| $B$ | $0,0,0$ | $3,2,1$ |

G2

Find all Nash Equilibria of the following Games:

- Game 1:
  $(B, R)$ is the only pure NE since there is no unilateral deviation that is profitable with payoff $(1, 1)$ (in the exam must explain explicitly in detail why it is a mutual BR)
  $\{(0, 5T + 0.5M, 0.5L + 0.5C), (0, 25T + 0.25M + 0.5B, 0.25L + 0.25C + 0.5R)\}$ with payoffs $(1, 1)$ and $(0.5, 0.5)$ respectively are the mixed NE

- Game 2:
  $\{(T, L), (T, C), (B, R)\}$ are the pure NE with $(1, 1)$ as payoff.
  $\{(05T+0.5B, 0.5L+0.5R), (0.5T+0.5B, 0.5C+0.5R)\}$ with payoffs $(0.5, 0.5)$ and $(0.5, 0.5)$ respectively
  $(T, pL + (1-p)C)$ where $p \in (0, 1)$ with payoffs $(1, 1)$ and $(0.5T + 0.5B, qL + (1/2-q)C + 0.5R)$ where $q \in (0, 0.5)$ with payoffs $(0.5, 0.5)$

- Game 3:
  $\{(T, L, G1), (B, R, G2)\}$ are pure NE
  $\{(0.25T + 0.75B, 0.5L + 0.5R, 0.75G1 + 0.25G2)\}$ is the unique mixed NE with payoffs $(3/8, 3/8, 3/8)$.
  Suppose P1 mixes $(\alpha, 1-\alpha)$ over $(T, B)$, P2 mixes $(\beta, 1-\beta)$ over $(L, R)$ and $(\gamma, 1-\gamma)$ over $(G1, G2)$.
  Then we have

  $$\begin{aligned} u_1(T) &= \beta_1\gamma_1, u_1(B) = 3(1-\beta)(1-\gamma) \\ u_2(L) &= 2\alpha\gamma, u_2(R) = 2(1-\alpha)(1-\gamma) \\ u_3(G1) &= 3\alpha\beta, u_3(G2) = (1-\alpha)(1-\beta) \end{aligned}$$

  which gives

  $$\begin{aligned} \beta\gamma &= 3(1-\beta)(1-\gamma) \\ \alpha\gamma &= (1-\alpha)(1-\gamma) \\ 3\alpha\beta &= (1-\alpha)(1-\beta) \end{aligned}$$

  Solving yields

  $$\alpha = 1/4, \beta = 1/2, \gamma = 3/4$$

**Example 1.24 (Games with Continuous Actions (Bertrand Game)).** In the Bertrand Game we have

- Players: 2 firms
- Actions: $A_i = \mathbb{R}_+$ where the firm chooses $p_i \in A_i$
- Payoff: a function $A_1 \times A_2 \to \mathbb{R}$ i.e.

$$\pi_1(p_1, p_2) = \begin{cases} (p_i - c)Q(p_i) & \text{if } p_i < p_{-i} \\ 0.5(p_i - c)Q(p_i) & \text{if } p_i = p_{-i} \\ 0 & \text{if } p_i > p_{-i} \end{cases}$$

Denote

$$p^m = \underset{p}{\operatorname{argmax}}(p - c)Q(p)$$

Consider firm $i$:

- $p_{-i} < c$: negative profits if we win competition or share, zero if lose. Thus,

$$B_i(p_{-i}) = \{p_i : p_i > p_{-i}\}$$

- $p_{-i}$: negative profits if we win competition, zero if we loose or share. Thus,

$$B_i(p_{-i}) = \{p_i : p_i \geq p_{-i}\}$$

- $c < p_{-i} \leq p^m$ then $B_i(p_{-i}) = \emptyset$.
  - Firm $i$ should charge $c < p_i \leq p_{-i}$ since in this interval $\pi_i(p_i, p_{-i}) > 0$ and for $p_i \leq c$ or $p_i > p_{-i}$ we have $\pi_i(p_i, p_{-i}) \leq 0$
  - Assume there is a best response $p_i^* < p_{-i}$ i.e. $\pi_i(p_i^*, p_{-i}) \geq \pi_i(p_i, p_{-i})$ for all $p_i$. Then let us define $\varepsilon = p_{-i} - p_i^* > 0$. Consider another price $\tilde{p} = p_{-i} - \varepsilon/2$. Then

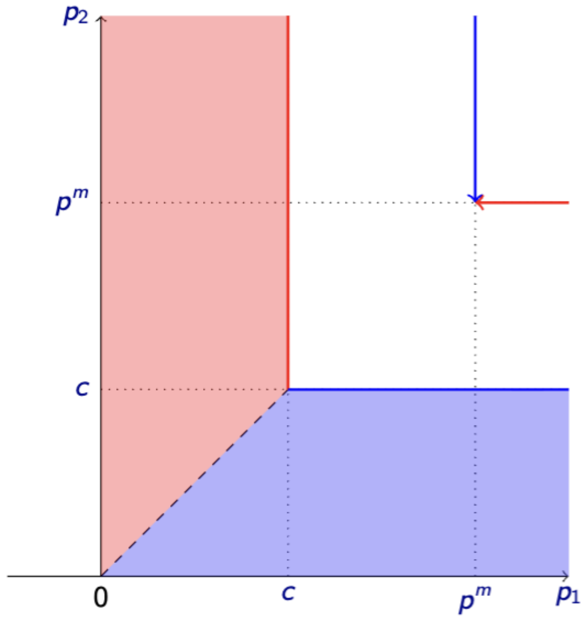
    $$\pi_i(\tilde{p}, p_{-i}) > \pi(p_i^*, p_{-i})$$

    Thus $p_i^*$ is not the best response
  - Assume there is an optimal response $p_i^* = p_{-i}$. Then there exists $\varepsilon > 0$ such that $\pi(p_I^* - \varepsilon, p_{-i}) > \pi(p_i^*, p_{-i})$ thus $p_i^*$ is not the best response.
- $p_{-i} > p^m$, just charge $B_i(p_{-i}) = p^m$ since it wins for sure and maximises profits.

Therefore,

$$B_i(p_{-i}) = \begin{cases} \{p_i : p_i > p_{-i} & \text{if } p_{-i} < c \\ \{p_i : p_i \geq p_{-i}\} & \text{if } p_{-i} = c \\ & \text{if } c < p_{-i} \leq p^m \\ \{p^m\} & \text{if } p_m < p_{-i} \end{cases}$$

Where the graph is




Note that this is a correspondence not a function. Thus

$$p_1 = p_2 = c$$

is the unique equilibrium

**Definition 1.25 (Mixed Nash Equilibrium).** A strategy profile $\alpha^* \in_{\times j \in N} \Delta(A_j)$ is an NE (in mixed strategies) if for any $\alpha_i \in \Delta(A_i)$ of player $i$

$$v_i(\alpha_i^*, \alpha_{-i}^*) \geq v_i(\alpha_i, \alpha_{-i}^*)$$

Similarly

**Proposition 1.26.** The set of Nash Equilibrium coincides with the intersection of Best Response correspondences. Formally, the action profile $a_i^*$ is a nash equilibrium if and only if

$$\alpha_i^* \in B_i(\alpha_{-i}^*), \forall i \in N$$

**Theorem 1.27 (Nash's Theorem).** Every finite strategic game has a mixed Nash Equilibrium

*Proof.* (Outline) The proof uses Kakutani's Fixed Point Theorem for Correspondences to show that every BR correspondence has a fixed point (which is the mixed NE) ∎

To find a mixed nash equilibrium, we need to find probabilities over player $i$'s actions such that players $-i$ are indifferent over their actions

**Proposition 1.28 (Indifference Principle).** The mixed strategy profile $\alpha^*$ is NE iff for each player $i$

- $\alpha_i$ assigns zero probability to all pure actions $a_i$ for which $v_i(a_i, \alpha_{-i}^*) < v_i(\alpha_i^*, \alpha_{-i}^*)$
- there are no actions $a_i$ for which $v_i(a_i, \alpha_{-i}^*) > v_i(\alpha_i^*, \alpha_{-i}^*)$

*Proof.* The idea is that if there is $a_i$ played with positive probability for which $v_i(a_i, \alpha_{-i}^*) < v_i(\alpha_i^*, \alpha_{-i}^*)$, then a player is better off relocating probability mass to actions with higher expected payoff (profitable deviation). If there is $a_i$ played with zero probability for which $v_i(a_i, \alpha_{-i}^*) > v_i(\alpha_i^*, \alpha_{-i}^*)$ then a player is better off relocating probability mass to this action. ∎

**Remark.** Notice that this says that strictly dominated strategies do not feature in mixed strategies

**Example 1.29 (Mixed NE with Finite Actions).**

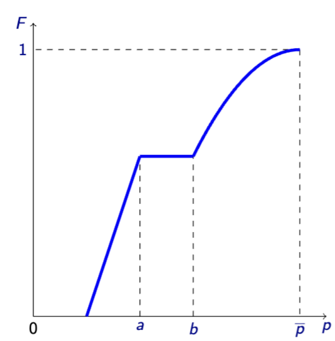
We often look at symmetric mixed equilibrium with the same cdf $F(z)$. We often have to prove certain properties of the cdf such as

- The support of $F(z)$ is a convex set i.e. there is no gap in any interval $[a, b]$
- $F(z)$ is continuous i.e. there is no atom at $z_0$ in the support where there is a mass of probability

Then we need to either find the upper or lower bound of the support and use the property of indifference to solve for the CDF (which is the mixed strategy)

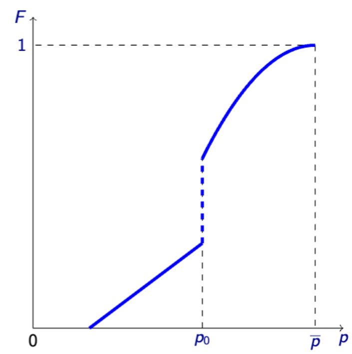
**Example 1.30 (Mixed NE with Continuous Actions I).** Consider two firs producing an homogenous good at marginal production cost $c$ where there is a unit mass of consumers where each consumer has a valuation $v > c$ for the good (same for all consumers). The fraction $\lambda$ of consumers are informed. These consumers know both the prices and buy at the lowest price. If the prices are equal, they split evenly. Fraction $1 - \lambda$ of consumers are uninformed and they just buy the good at the closest store and we assume they split equally across the firms. We want to find the equilibrium prices in the model (assuming only symmetric mixed strategies)
*Solution*: No pure strategies. Consider $p_1 = p_2 = c$ as an equilibrium. But one firm can deviate by charging $p_i = v$ as uninformed still buy from this firm so its profits equal $\frac{1-\lambda}{2}(v - c) > 0$ (initially their profits are 0). Consider $p_1 = p_2 = p > c$ as an equilibrium, but then one firm can deviate by charging $p_1 = p - \varepsilon$. Then all informed move to this firm so it gains $\lambda(p_1 - c)/2$ and loses less than $\varepsilon$ so it is a profitable deviation.
Define the mixed strategy of firm $i$ be the cdf $F_i(p) = P(\text{pricecharged} \leq p)$ and we only look at a symmetric candidate equilibrium where both firms play $F(p)$ i.e. the same mixed strategy.
The key idea is to examine the properties of the support of $F$ and to find either the lower or upper bound and apply indifference

1. Claim 1: the support of $F(p)$ is a convex set. If there is a gap $[a, b]$ in the support, then $\pi(a) < \pi(b-\varepsilon)$ for sufficiently small $\varepsilon$.


   This a contradiction because under indifference, no action should give you a higher payoff

2. Claim 2: $F(p)$ is continuous. If there is an atom at $p_0$, then $\pi(p_0 - \varepsilon) > \pi(p_0 + \varepsilon)$ i.e. there is a gap above $p_0$.




But again this violates indifference.

3. The expected profit is
$$\pi(p) = \lambda[1 - F(p)](p - c) + \frac{1-\lambda}{2}(p - c)$$
the firm always gets the second term from uninformed and gets the informed only if the sell the lowest price which happens with probability $1 - F(p)$ since $F(p) = P(\text{pricecharged} \leq p)$

4. Claim 3: upper bound of the support $\bar{p} = v$. Note that at $\bar{p}$, a firm sells only to loyal customers and earns $\frac{1-\lambda}{2}(\bar{p} - c)$. Thus it can deviate by setting $v$ and earning a higher profit.

5. Using the boundary condition $\bar{p} = v$ we have
$$\pi = \frac{1-\lambda}{2}(v - c)$$
by the indifference principle, the expected profits are constant for any $p$ which yields
$$\pi(p) = \lambda[1 - F(p)](p - c) + \frac{1-\lambda}{2}(p - c) = \frac{1-\lambda}{2}(v - c)$$
giving us
$$F(p) = 1 - \left(\frac{1-\lambda}{2\lambda}\frac{v-p}{p-c}\right)$$
The lower bound of the support is defined using $F(\underline{p}) = 0$ which yields
$$\underline{p} = \frac{(1-\lambda)v + 2\lambda c}{1+\lambda}$$

**Example 1.31 (Mixed NE with Continuous Actions II).** Two 'oligarchs' $A$ and $B$ have a dispute over an asset of value $X$. Although both oligarchs now both reside in city $L$, they decided to settle their dispute in court in city $M$, where they originally made their money. The reason is that the legal system in city $M$ is simple and transparent: both contestants provide the judge with the 'evidence', which takes the form of a briefcase with cash. The judge keeps the evidence from both contestants and decides the case in favour of the one who provided more evidence. If both contestants provided an equal amount of evidence the judge flips a fair coin to make a decision. Assume that oligarchs submit their briefcases simultaneously and any contribution $y \in \mathbb{R}^+$ can be put in the envelope.

1. Derive the best response of oligarch $A$ to oligarch's $B$ contribution
*Solution*: The utility functions are
$$u_A(y_a, y_b) = \begin{cases} X - y_a, & \text{if } y_a > y_b \\ \frac{1}{2}X - y_a, & \text{if } y_a = y_b \\ -y_a, & \text{if } y_a < y_b \end{cases}$$

and

$$u_B(y_a, y_b) = \begin{cases} X - y_b, & \text{if } y_b > y_a \\ \frac{1}{2}X - y_b, & \text{if } y_a = y_b \\ -y_b, & \text{if } y_b < y_a \end{cases}$$

Consider $A$'s BR to $B$'s contribution for $y_a = b \geq 0$. If $b \geq X$, then $A$s payoff is

$$u_A(y_a, y_b) = \begin{cases} X - y_a < 0, & \text{if } y_a > b \\ \frac{1}{2}X - y_a < 0, & \text{if } y_a = 0 \\ -y_a \leq 0, & \text{if } y_a < b \end{cases}$$

So $A$'s BR is $y_a = 0$. If $b < X$, then $A$'s BR is $y_a = b + \varepsilon$ for $\varepsilon > 0$

2. Using your answer to part (1) show that there is no pure strategy equilibrium.
*Solution*: Want to show that there is no $(y_a^*, y_b^*)$ which is an NE. Consider by dichotomy,
Case 1: Suppose $y_b^* < X$, then $y_a^* = y_b^* + \varepsilon$. But if $y_a^* = y_b^* + \varepsilon < X$, there is a profitable deviation for $y_b = y_a^* + \eta$ with $\eta > 0$ so that $u_b = X - y_b^* - \varepsilon - \eta > 0 > -y_b^*$. If $y_a^* = y_b^* + \varepsilon = X$, then there is a profitable deviation for $y_b = 0$ so that $u_b = 0 > -y_b^*$.
Case 2: Suppose $y_b^* \geq X$, then $y_a^* = 0$ and so there is a profitable deviation so that $y_b = y_b^* - \varepsilon$ for $\varepsilon > 0$ so that $u_b = X - y_b^* + \varepsilon > X - y_b^*$.
Hence, no pure NE

3. Prove that there is no contribution value played with positive probability.
*Solution*: Suppose, for contradiction, that oligarch $B$'s mixed strategy has an atom at some $a \in \mathbb{R}_+$, i.e.

$$\Pr(y_B = a) = p > 0.$$

Let $F_B(t-) := \Pr(y_B < t)$ denote the left limit of $B$'s CDF at $t$.

If $A$ bids $a$, then $A$ wins when $y_B < a$ and ties when $y_B = a$ (winning the tie with probability 1/2), so

$$U_A(a) = X\, F_B(a-) + \frac{X}{2}p - a.$$

If instead $A$ bids $a + \varepsilon$ for some small $\varepsilon > 0$, then $A$ wins whenever $y_B < a + \varepsilon$, so

$$U_A(a + \varepsilon) = X\, F_B(a + \varepsilon-) - (a + \varepsilon).$$

Therefore,

$$\begin{aligned} U_A(a+\varepsilon) - U_A(a) &= X\big(F_B(a+\varepsilon-) - F_B(a-)\big) - \varepsilon - \frac{X}{2}p \\ &= X\Big(\Pr(a \leq y_B < a+\varepsilon)\Big) - \varepsilon - \frac{X}{2}p \\ &= X\Big(p + \Pr(a < y_B < a+\varepsilon)\Big) - \varepsilon - \frac{X}{2}p \\ &\geq \frac{X}{2}p - \varepsilon. \end{aligned}$$

Choose $\varepsilon < \frac{X}{2}p$. Then $U_A(a+\varepsilon) > U_A(a)$, so $A$ has a profitable deviation, contradicting that $B$'s strategy is part of a mixed Nash equilibrium.

Hence $p = 0$ for every $a$, i.e. there is no contribution level played with positive probability.

4. Using your answer to (3), or otherwise, show if $F(y)$ is a symmetric mixed strategy equilibrium distribution then it does not have gaps in its support.
*Solution*: Suppose $A$ and $B$ play symmetric mixed strategies $F(y_a)$ and $F(y_b)$. By (3), $A$'s payoff is

$$u_A(y, y_b) = XF(y_b < y) - a$$

Suppose there is some gap $(a, b)$ with $0 \le a < b$. Then,

$$F(y_b < a) = F(y_b < b)$$

i.e. you increase how much you pay without winning anythng. But then, there is a profitable deviation s.t. $A$ will always bid $a$ since

$$u_A(a, y_b) = XF(y_b < a) - a > u_A(b, y_b) = XF(y_b < b) - b$$

since $a < b$ by construction. Since $F(y_a)$ and $F(y_b)$ is a symmetric mixed equilibrium, by indifference

$$u_A(a, y_b) = u_A(b, y_b)$$

a contradiction

5. Argue that the lowest contribution in the support of $F(y)$ must equal to zero. *Solution*: Let the support of the symmetric mixed equilibrium strategy be $S = [\underline{y}, \bar{y}]$. Suppose for contradiction that $\underline{y}_b > 0$. Then $B$ never bids below $\underline{y}_b$. Then, $u_A(y_a, y_b) = XF(y_b < y_a) - y_a$ by (3) since there is no atom, then $y_a = a$ for $a < y_b$ s.t. $F(y_b < a) = 1$. Hence,

$$u_A(a, y_b) = X - a > u_A(y_a, y_b) = XF(y_b < y_a) - y_a$$

so there is a profitable deviation since $0 < F(y_b < y_a) \le 1$ for $y_a > a$ and so a mixed equilibrium cannot be sustained. Define $a = \underline{y}_b - \varepsilon$ for $\varepsilon > 0$. Then for any choice $\underline{y}_b > 0$, $\exists a \in \mathbb{R}_+$ s.t. the above holds and so $\underline{y}_b = 0$ and by symmetry, $\underline{y}_a = 0$. Therefore, $\underline{y} = 0$

6. Derive the equilibrium strategy of oligarchs.
*Solution*: By (3), consider

$$u_A(y_a, y_b) = XF(y_b < y_a) - y_a$$

By (5), $\underline{y}_b = 0$ so that $F(0) = 0$. Then, $u_A(0, y_b) = XF(y_b < 0) - 0 = 0$. By indifference, $u_A(0, y_b) = u_A(y_a, y_b) = XF(y_b < y_a) - y_a = 0$. So,

$$F(y_b < y_a) = \frac{y_a}{X} \text{ and } F(y_a < y_b) = \frac{y_b}{X}$$

with support $S \in [0, X]$

**Discussion on NE, Rationalizability and IESDS**

**Proposition 1.32.** In finite games the set of correlatedly rationalizable actions coincides with the set of actions surviving the iterated elimination of strictly dominated strategies.
In finite games with two players, the set of rationalizable actions coincides with the set of actions surviving the iterated elimination of strictly dominated strategies.

The set of Nash Equilibria are a subset of rationalizable actions

Nash Equilibrium needs rationality (but not rationalizability in the sense of requiring common knowledge of rationality) and correct anticipation of what the other does; but rationalizability only needs common knowledge of rationality and no need correct anticipation of what the other does.

## 1.4 Correlated Equilibrium

**Definition 1.33 (Correlated Equilibrium).** A correlated equilibrium of a strategic game consists of

- a finite state space $\Omega$ and a probability meaure $\mu$ over this space
- for each player $i$, a partition $\mathcal{P}_i(\omega)$

- for each player $i$ a strategy $\sigma_i : \Omega \to A_i$ with $\sigma_i(\omega) = \sigma_i(\omega')$ whenever $\omega, \omega' \in P_i$ wih $P_i \in \mathcal{P}_i$ such that for all $i$ and $\omega$

$$\sum_{\omega' \in \Omega} \mu(\omega' | P_i(\omega)) u_i(\sigma_i(\omega'), \sigma_{-i}(\omega')) \geq \sum_{\omega' \in \Omega} \mu(\omega' | P_i(\omega)) u_i(s_i(\omega'), \sigma_{-i}(\omega'))$$

**Example 1.34.** Consider the strategic form game represented by the following matrix

| | $L$ | $R$ |
|---|---|---|
| $T$ | $6,6$ | $2,8$ |
| $B$ | $8,2$ | $0,0$ |

1. What are the Nash equilibria of this game and what are the player's expected payoffs in each equilibrium?
   *Solution*: Pure NE: $\{(T,R),(B,L)\}$ with payoffs $(2,8),(8,2)$ respectively. Consider a mixed strategy of $(p, 1-p)$ over $(T,B)$ and $(q, 1-q)$ over (L,R). Then

$$u_1(T) = 6q + 2(1-q) = 8q = u_1(B)$$

   yielding

$$q = 0.5$$

   An analogous argument yields $p = 0.5$. So the NE is $(0.5T + 0.5B, 0.5L + 0.5R)$, with expected payoffs $(4,4)$.
   Suppose a single fair coin is tossed, the outcome of which is publicly observed. If it is heads player 1 is told to play $T$ and player 2 is told to play $R$; if it is tails, player 1 is told to play $B$ and player 2 is told to play $L$

2. Would each player be willing to follow these instructions, given that the other player did so? What would be the expected payoffs?
   *Solution*: If $\{T\}$, then $(8,2)$ and if $\{H\}$, then $(2,8)$. Expected payoff is $(5,5)$. If tails, then since P1 expects P2 to play $L$, deviating to $T$ reduces payoff from 8 to 6. If heads, since P1 expects P2 to play $R$, deviating to $B$ reduces payoffs from 2 to 0. So P1 has incentives to follow. Likewise, P2 would have the incentives to follow. The public signal allows players to randomise over the two pure NE to yield a higher payoff.
   A "three sided coin" is tossed with equal probability assigned to each of the three outcomes: $x, y$ and $z$. The player do not observe the outcome. However, if $x$ is thrown, player 1 is told to play $T$ and player 2 is told to play $R$. If $y$ is thrown, player 1 is told to play $T$ and player 2 is told to play $L$. If $z$ is thrown, player 1 is told to play $B$ and player 2 is told to play L.

3. Show that each player would be willing to follow instructions given that the other player does so. What would be the expected payoffs? Comment.
   *Solution*: We have

$$\begin{aligned} \{x\} &\implies (T,R) \text{ with } (2,8) \\ \{y\} &\implies (T,L) \text{ with } (6,6) \\ \{z\} &\implies (B,L) \text{ with } (8,2) \end{aligned}$$

   Suppose P1 is told to play $B$ which happens only under $\{z\}$. Then P1 could not profitably deviate since he expects P2 to play $L$ and 8 when playing $T$ is the highest payoff he can receive in the game. Suppose P1 is told to play $T$ which happens under $\{x, y\}$. Since $x, y$ occur with $p_x = p_y = 1/3$, $P(x|T) = P(y|T) = 1/2$. Hence

$$u_1(T) = 1/2(2) + 1/2(6) = 4$$

   If P1 deviates to $B$,

$$u_1(B) = 1/2(8) + 0 = 4$$

   P1 is indifferent and hence willing to follow. We can construct an analogous argument to show that P2 is willing to follow. Expected payoffs are $(16/3, 16/3)$ which is even higher.

**Remark.** There is a public correlating device that can allow players to mix across Nash Equilibria – but the correlating device must be incentive compatible. Furthermore, it is without loss of generality to restrict attention to restrict attention to correlating devices over strategy sets, instead of trying to identify all correlated equilibria for a given set of playoffs (since the correlated equilibrium of a game depends on the information structure)

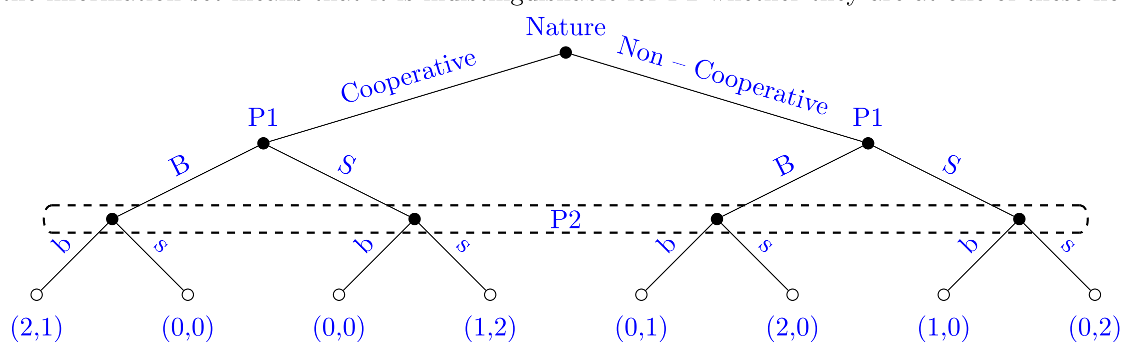
# 2 Static Games with Incomplete Information

## 2.1 Incomplete Information and Bayes-Nash Equilibrium

**Definition 2.1 (Game of incomplete information).**

We can think of a game with incomplete information as a game that begins with a move of the artificial player Nature and where there are information sets which reflect private information of players types known only to themselves.

**Example 2.2.** Consider the BoS game where P1 can either be Cooperative or Non-cooperative. Note that the information set means that it is indistinguishable for P2 whether they are at one of these nodes.




**Definition 2.3 (Bayes-Nash Equilibrium).** A Nash equilibrium of a Bayesian game (Bayes-Nash equilibrium) is a Nash equilibrium of a strategic game defined as follows:

- Players: Set of all pairs $(i, s_i)$ where $i$ is a player of a Bayesian game and $s_i$ is a signal that player $i$ can receive
- Actions: Set of actions of player $(i, s_i)$ is a set of actions of player $i$ in the Bayesian game
- Payoffs: Bernoulli payoff function of player $(i, s_i)$:

$$u_{i,s_i}(a_i, a_{-i}(\omega)) = \int_{\omega\in\Omega} u_i(a_i, a_{-i}(s_{-i}(\omega)), \omega) d\mathbb{P}(\omega|s_i)$$

  where $a_{-i}(s_{_i}(\omega))$ is an action profile of all other players given the signals they might get in state $\omega$: $a_j(j, s_j(\omega))$

A (pure) *Bayes–Nash equilibrium* is a strategy profile $a^* = (a_i^*)_{i\in I}$ with $a_i^* : S_i \to A_i$ such that for every player $i$ and every type $t \in S_i$ with $\mathbb{P}(s_i = t) > 0$,

$$a_i^*(t) \in \arg\max_{a_i\in A_i} \mathbb{E}\big[u_i\big(a_i,\, a_{-i}^*(s_{-i}(\omega)),\, \omega\big) \mid s_i(\omega) = t\big]$$

Note that the above definition uses interim payoffs (after learning one's type but before learning the type of all others) rather than ex-ante payoffs (before learning one's type). The ex-ante Bayes-Nash maximises

$$E\big[u_i\big(a_i,\, a_{-i}^*(s_{-i}(\omega)),\, \omega\big)\big]$$

**Theorem 2.4.** An action profile $a^* = (a_i^*)_{i\in I}$ is a interim Bayes-Nash equilibrium iff it is an ex-ante Bayes-Nash equilibrium

In other words, it does not matter which they compute.

**Remark.** However, it is very important to note that this equivalence does not hold for IESDS

We will illustrate the equivalence of BNE in the next example.

**Example 2.5 (BNE in finite games PSet3Q2).** Consider a penalty game between a striker and a goalkeeper. The striker can shoot left or right corner, the goal keeper can jump left or right (from the striker side). The striker can be of two types: with a preference for left corner ('Left type') or with a preference for right corner ('Right type'). The goalkeeper does not know the striker type and thinks that both types are equally likely. Assuming that the striker is the row player and the goalkeeper is the column player the payoffs are:

| $St/G$ | $l$ | $r$ |
|---|---|---|
| $L$ | $0.7, 0.3$ | $1, 0$ |
| $R$ | $0.8, 0.2$ | $0.6, 0.4$ |

Left Type

| $St/G$ | $l$ | $r$ |
|---|---|---|
| $L$ | $0.6, 0.4$ | $0.8, 0.2$ |
| $R$ | $1, 0$ | $0.7, 0.3$ |

Right Type

*Solution*: First, we want to represent the game both in terms of *interim* and *ex ante* payoffs, and show that the results don't change.
**Ex-ante Payoff Representation**
We may represent the ex-ante payoffs in a single matrix where the striker does not know his type

| $St/G$ | $l$ | $r$ |
|---|---|---|
| $LL'$ | $6.5, 3.5$ | $9, 1$ |
| $LR'$ | $8.5, 1.5$ | $8.5, 1.5$ |
| $RL'$ | $7, 3$ | $7, 3$ |
| $RR'$ | $9, 1$ | $6.5, 3.5$ |

**Interim Payoff Representation**: We have

| $St/G$ | $l$ | $r$ |
|---|---|---|
| $L$ | $(0.7, 0.6), 0.35$ | $(1, 0.8), 0.1$ |
| $R$ | $(0.8, 1), 0.1$ | $(0.6, 0.7), 0.35$ |

in both cases, we computed the average payoff for the goalkeeper since he never learns the type of the striker. In both cases, we yield that there is no pure Bayes-Nash since the best responses are

$$BR_S(l) = \begin{cases} R & \text{if left type} \\ R & \text{if right type} \end{cases} \quad BR_S(r) = \begin{cases} L & \text{if right type} \\ L & \text{if right type} \end{cases}$$

and

$$BR_G(L) = l, \quad BR_G(R) = r$$

Now, we find the equilibria in which each type of pure striker plays a pure strategy with the goalkeeper playing a mixed strategy.
Then, if $S$ plays $(L, L)$, $G$ prefers $l$ strictly so no mix; if $S$ plays $(R, R)$, $g$ prefers $r$ strictly so no mix. If $S$ plays $L, R)$, then

$$u_G(l) = \frac{1}{2}(0.3) = u_G(r) = \frac{1}{2}(0.3) = 0.15$$

so $G$ is indifferent and will mix. Want to check if $(L, R)$ is a best response to $G$'s mixed strategy of playing $(l, r)$ w.p. $(p, 1-p)$.
If we consider interim payoffs:

- For left type: $u_S(L) = 0.7p + (1-p) = 1 - 0.3p$. Also, $u_S(R) = 0.8p + (1-p)0.6 = 0.6 + 0.2p$. $L$ is BR for left type iff $1 - 0.3p \geq 0.6 + 0.2p$ which yields $p \leq 0.8$.

- For right type: $u_S(L) = 0.6p + 0.8(1-p) = 0.8 - 0.2p$. Also, $u_S(R) = p + 0.7(1-p) = 0.7 + 0.3p$. $R$ is BR for right type iff $0.7 + 0.3p \geq 0.8 - 0.2p$ which implies $p \geq 0.2$.

So, the NE is $\{(L, R), pl + (1-p)r\}$ where $p \in [0.2, 0.8]$.
If we consider ex-ante payoffs, we just have to check that $LR'$ is the best response to $G$'s mixed strategy. Notice that $RL'$ is strictly dominated so we do not have to consider that. We have

$$8.5 \geq 6.5p + 9(1-p) \implies p \geq 0.2$$

and

$$8.5 \geq 9p + 6.5(1-p) \implies g \leq 0.8$$

which gives us the same set of NE.
If $S$ plays $(R, L)$, then

$$u_G(l) = 0.5(0.2) + 0.5(0.4) = u_G(r) = 0.5(0.4) + 0.5(0.2) = 0.3$$

so $G$ will mix. Suppose $G$ mixes over $(l, r)$ with $(q, 1-q)$, then wts $(R, L)$ is BR for $S$ (we only show interim payoffs).

- For left type, we have

$$u_S(L) = 0.7q + 1 - q = 1 - 0.3q$$
$$u_S(R) = 0.8q + 0.6(1-q) = 0.6 + 0.2q$$

  so $R$ is a best response iff $0.6 + 0.2q \geq 1 - 0.3q \implies q \geq 0.8$

- For right type,

$$u_S(L) = 0.6q + 0.8(1-q) = 0.8 - 0.2q$$
$$u_S(R) = q + 0.7(1-q) = 0.7 + 0.3q$$

  so $L$ is a best response iff $0.8 - 0.2q \geq 0.7 + 0.3q \implies q \leq 0.2$

which is impossible.
We want to see if either left or right type plays a mixed strategy (and show that this is impossible). We consider cases where goalkeeper mixes. Consider left type

$$u_S(L) = 0.7p + (1-p) = 1 - 0.3p$$
$$u_S(R) = 0.8p + (1-p)0.6 = 0.6 + 0.2p$$

So, $L \sim R \iff 1 - 0.3p = 0.6 + 0.2p \iff p = 0.8$. Consider right type

$$u_S(L) = 0.6 + 0.8(1-p) = 0.8 - 0.2p$$
$$u_S(R) = p + 0.7(1-p) = 0.7 + 0.3p$$

So, $L \sim R \iff 0.8 - 0.2p = 0.7 + 0.3p \iff p = 0.2$, which is impossible so both types cannot simultaneously mix. Now,

- Consider only Left type mixes: then, $p = 0.8$ the Right type's expected payoffs are

$$u_S(L) = 0.8 - 0.2(0.8) = 0.64, \qquad u_S(R) = 0.7 + 0.3(0.8) = 0.94,$$

  so the Right type plays $R$ purely (i.e. $p_R = 0$).

  Now compute the goalkeeper's expected payoff from $l$ versus $r$ (using prior 1/2 on each type).

  If the goalkeeper plays $l$, its expected payoff is

$$u_G(l) = \frac{1}{2}\Big(0.3p_L + 0.2(1-p_L)\Big) + \frac{1}{2}(0) = 0.1 + 0.05p_L.$$

  If the goalkeeper plays $r$, its expected payoff is

$$u_G(r) = \frac{1}{2}\Big(0 \cdot p_L + 0.4(1-p_L)\Big) + \frac{1}{2}(0.3) = 0.35 - 0.2p_L.$$

Therefore

$$u_G(r) - u_G(l) = \big(0.35 - 0.2p_L\big) - \big(0.1 + 0.05p_L\big) = 0.25(1 - p_L).$$

If the Left type mixes, then $p_L \in (0, 1)$, so $1 - p_L > 0$ and hence

$$u_G(r) > u_G(l).$$

Thus the goalkeeper's best response is to play $r$ purely (i.e. $q = 0$), contradicting the requirement $q = 0.8$ for Left-type mixing. So no BNE exists with Left type mixing.

- Assume the Right type mixes. Then $q = 0.2$.

  Given $q = 0.2$, the Left type's expected payoffs are

  $$u_S(L) = 1 - 0.3(0.2) = 0.94, \qquad u_S(R) = 0.6 + 0.2(0.2) = 0.64,$$

  so the Left type plays $L$ purely (i.e. $p_L = 1$).

  Again compute the goalkeeper's expected payoff.

  If the goalkeeper plays $l$:

  $$u_G(l) = \frac{1}{2}(0.3) + \frac{1}{2}\Big(0.4p_R + 0 \cdot (1 - p_R)\Big) = 0.15 + 0.2p_R.$$

  If the goalkeeper plays $r$:

  $$u_G(r) = \frac{1}{2}(0) + \frac{1}{2}\Big(0.2p_R + 0.3(1 - p_R)\Big) = 0.15 - 0.05p_R.$$

  Hence

  $$u_G(l) - u_G(r) = 0.25p_R.$$

  If the Right type mixes, then $p_R \in (0, 1)$, so $p_R > 0$ and thus

  $$u_G(l) > u_G(r).$$

  So the goalkeeper's best response is to play $l$ purely (i.e. $q = 1$), contradicting the requirement $q = 0.2$ for Right-type mixing. So no BNE exists with Right type mixing.

**Example 2.6 (BNE in continuous games PSet2Q4).** Consider a Cournot duopoly game. Two firms compete in quantities in a market with demand function $p = 1 - q_1 - q_2$. Suppose that each firm has a production cost $c \in \{c_L, c_H\}$, which is the firm's private information. Suppose, then, the joint distribution of costs is defined as

| Firm 1/Firm 2 | $c_L$ | $c_H$ |
|---|---|---|
| $c_L$ | $\alpha$ | $\frac{1}{2} - \alpha$ |
| $c_H$ | $\frac{1}{2} - \alpha$ | $\alpha$ |

where $\alpha \in [0, \frac{1}{2}]$. That is, both costs are ex ante equally likely and parameter $\alpha$ captures the correlation between the costs. Assume that costs are small enough so that the outputs are positive.

1. Suppose that firm $i \in \{1, 2\}$ has cost realisation $c_j, j \in \{L, H\}$. What is the posterior belief of firm $i$ about firm $-i$'s costs?

   From the joint distribution,

   $$\Pr(c_1 = c_L, c_2 = c_L) = \alpha, \quad \Pr(c_1 = c_L, c_2 = c_H) = \tfrac{1}{2} - \alpha,$$

   $$\Pr(c_1 = c_H, c_2 = c_L) = \tfrac{1}{2} - \alpha, \quad \Pr(c_1 = c_H, c_2 = c_H) = \alpha.$$

   Hence, using Bayes' rule,

   $$\Pr(c_{-i} = c_L \mid c_i = c_L) = \frac{\alpha}{\alpha + (\frac{1}{2} - \alpha)} = 2\alpha, \qquad \Pr(c_{-i} = c_H \mid c_i = c_L) = 1 - 2\alpha,$$

   $$\Pr(c_{-i} = c_H \mid c_i = c_H) = \frac{\alpha}{\alpha + (\frac{1}{2} - \alpha)} = 2\alpha, \qquad \Pr(c_{-i} = c_L \mid c_i = c_H) = 1 - 2\alpha.$$

2. Define the strategy of each player and set up the payoff function for each player
   A (pure) strategy for firm $i$ is a mapping

$$s_i : \{c_L, c_H\} \to \mathbb{R}_+, \qquad s_i(c_L) = q_{iL}, \quad s_i(c_H) = q_{iH}.$$

   Given quantities $(q_1, q_2)$ the price is $p = 1 - q_1 - q_2$ and profit is

$$\pi_i = (p - c_i)q_i = (1 - q_1 - q_2 - c_i)q_i.$$

   Therefore the interim expected payoff of firm 1 is

$$\begin{aligned}
\Pi_{1H}(q_{1H}, q_{2H}, q_{2L}) &= \Big(2\alpha(1 - q_{1H} - q_{2H}) + (1-2\alpha)(1 - q_{1H} - q_{2L}) - c_H\Big)q_{1H}, \\
\Pi_{1L}(q_{1L}, q_{2H}, q_{2L}) &= \Big((1-2\alpha)(1 - q_{1L} - q_{2H}) + 2\alpha(1 - q_{1L} - q_{2L}) - c_L\Big)q_{1L}.
\end{aligned}$$

   Firm 2's interim payoffs are symmetric:

$$\begin{aligned}
\Pi_{2H}(q_{2H}, q_{1H}, q_{1L}) &= \Big(2\alpha(1 - q_{2H} - q_{1H}) + (1-2\alpha)(1 - q_{2H} - q_{1L}) - c_H\Big)q_{2H}, \\
\Pi_{2L}(q_{2L}, q_{1H}, q_{1L}) &= \Big((1-2\alpha)(1 - q_{2L} - q_{1H}) + 2\alpha(1 - q_{2L} - q_{1L}) - c_L\Big)q_{2L}.
\end{aligned}$$

3. Compute the best response for each player and find the Bayes Nash Equilibrium of the game
   FOCs for firm 1 give

$$\begin{aligned}
\frac{\partial \Pi_{1H}}{\partial q_{1H}} = 0 &\iff q_{1H} = \frac{1 - c_H - \big(2\alpha q_{2H} + (1-2\alpha)q_{2L}\big)}{2}, \\
\frac{\partial \Pi_{1L}}{\partial q_{1L}} = 0 &\iff q_{1L} = \frac{1 - c_L - \big((1-2\alpha)q_{2H} + 2\alpha q_{2L}\big)}{2}.
\end{aligned}$$

   Similarly, firm 2's best responses are

$$\begin{aligned}
q_{2H} &= \frac{1 - c_H - \big(2\alpha q_{1H} + (1-2\alpha)q_{1L}\big)}{2}, \\
q_{2L} &= \frac{1 - c_L - \big((1-2\alpha)q_{1H} + 2\alpha q_{1L}\big)}{2}.
\end{aligned}$$

   In a symmetric BNE, $q_{1H} = q_{2H} = q_H$ and $q_{1L} = q_{2L} = q_L$. Solving the two linear equations yields

$$q_H^* = \frac{1 - 2c_H + c_L + 2\alpha(2 - c_H - c_L)}{3(1+4\alpha)}, \qquad q_L^* = \frac{1 - 2c_L + c_H + 2\alpha(2 - c_H - c_L)}{3(1+4\alpha)}.$$

   Thus the BNE strategy is: for each firm $i$,

$$s_i^*(c_H) = q_H^*, \qquad s_i^*(c_L) = q_L^*.$$

4. How does the output of each type of the player change with $\alpha$? Provide an intuitive explanation for this relation.
   Differentiate:

$$\frac{\partial q_H^*}{\partial \alpha} = \frac{2(c_H - c_L)}{(1+4\alpha)^2} > 0, \qquad \frac{\partial q_L^*}{\partial \alpha} = \frac{2(c_L - c_H)}{(1+4\alpha)^2} < 0 \quad (\text{since } c_H > c_L).$$

   *Intuition:* Higher $\alpha$ means costs are more positively correlated. If a firm learns it is low cost, it becomes more likely the rival is also low cost (tougher competition), so it produces less. If a firm learns it is high cost, it becomes more likely the rival is also high cost (softer competition), so it produces more.

We use symmetry here because the question is long and just “pick your battles"

## 2.2 Continuum of Types

Initially, we look at problems with discrete types; but there are problems with a continuum of player types that are distributed over a probability distribution. We usually look at linear equilibrium.

**Example 2.7 (Bertrand Competition).** Suppose that there is a single buyer who is willing to pay up to 1 pound for a product. There are two sellers, each seller has production cost $c_i$, which is independently uniformly distributed on $[0, 1]$. Sellers simultaneously make their offers and the buyer buys at the lowest price or randomly chooses the seller if the prices are equal. Find the BNE of the game.
We assume that the sellers follow a linear increasing pricing strategy which maps their type into price

$$p_i = \alpha_i + \beta_i c_i$$

We want to derive the equilibrium in linear pricing strategies i.e. find $\alpha_i$ and $\beta_i$. The expected profit of firm $i$ equals to (note that ties happen with probability zero):

$$\pi_i = (p_i - c_i) \times \mathbb{P}(p_i < p_{-i}) = (p_i - c_i) \times \mathbb{P}(p_i < \alpha_{-i} + \beta_{-i}c_{-i}) = (p_i - c_i) \times \mathbb{P}\left(\frac{p_i - \alpha_{-i}}{\beta_{-i}} < c_{-i}\right)$$

Thus we rewrite the profit function as

$$\pi_i = (p_i - c_i)\left(1 - \frac{p_i - \alpha_{-i}}{\beta_{-i}}\right)$$

Maximising wrt $p_i$ gives

$$p_i = \frac{1}{2}c_i + \frac{1}{2}(\alpha_{-i} + \beta_{-i})$$

Thus $\beta_i = 1/2$. Performing the same analysis for another firm we yield $\beta_{-i} = 1/2$. Thus,

$$\begin{aligned} p_1 &= \frac{1}{2}c_1 + \frac{1}{2}(\alpha_2 + \frac{1}{2}) \\ p_2 &= \frac{1}{2}c_2 + \frac{1}{2}(\alpha_1 + \frac{1}{2}) \end{aligned}$$

Equivalently,

$$\begin{cases} \alpha_1 = \frac{1}{2}(\frac{1}{2} + \alpha_2) \\ \alpha_2 = \frac{1}{2}(\frac{1}{2} + \alpha_1) \end{cases} \implies \alpha_1 = \alpha_2 = \frac{1}{2}$$

Thus the linear equilibrium is $p_i = \frac{c_i}{2} + \frac{1}{2}$

**Example 2.8 (Double Auction).** Consider the following problem

- A buyer has a valuation for the good $v_b \sim U[0, 1]$
- A seller has a valuation for the good $v_s \sim U[0, 1]$
- Both buyer and seller simultaneously announce their prices, $p_b$ and $p_s$
- If $p_b \geq p_s$, then the trade takes place at a price $(p_b + p_s)/2$, otherwise there is no trade
- Payoffs: 0 if $p_b < p_s$, otherwise

$$\pi_b(p_b, p_s, v_b) = v_b - \frac{p_b + p_s}{2} \quad \pi_s(p_b, p_s, v_b) = \frac{p_b + p_s}{2} - v_s$$

Suppose that the seller plays a linear strategy $p_s(v_s) = \alpha + \beta v_s$. The expected payoff of the buyer is

$$\pi_b = \mathbb{P}(\text{Trade}) \times [v_b - \text{ Expected Price}]$$

The first term is

$$\mathbb{P}(\text{Trade}) = \mathbb{P}[p_b \geq p_s] = \mathbb{P}[p_b \geq \alpha + \beta v_s] = \mathbb{P}\left[v_s \leq \frac{p_b - \alpha}{\beta}\right] = \frac{p_b - \alpha}{\beta}$$

Note that $p_s \sim U[\alpha, \alpha + \beta]$. Thus,

$$\text{Expected Price } = \frac{p_b + \mathbb{E}(p_s | p_s \leq p_b)}{2} = \frac{1}{2}\left(p_b + \frac{p_b + \alpha}{2}\right)$$

Hence,

$$\pi_b(p_b; v_b, \alpha, \beta) = \frac{p_b - \alpha}{\beta}\left[v_b - \frac{1}{2}\left(p_b + \frac{p_b + \alpha}{2}\right)\right] = \frac{1}{4\beta}(p_b - \alpha)(4v_b - 3p_b - \alpha)$$

The first order condition is

$$4v_b - 3p_b - \alpha - 3p_b + 3\alpha = 0 \implies p_b = \frac{2}{3}v_b + \frac{1}{3}\alpha$$

Now we can proceed by deriving the seller's best response assuming the buyer plays a linear strategy. Note that if we get the seller's best response is a linear strategy, then our assumption will be mutually consistent: a buyer's BR to a linear strategy of the seller is linear and vice versa.

Suppose that the buyer plays a linear strategy $p_b = \gamma + \delta v_b$. Then the payoff of the seller is given by

$$\pi_s = \mathbb{P}[\text{Trade}] \times [\text{Expected Price } - v_s] = \mathbb{P}(p_b \geq p_s) \times \left(\frac{p_s + \mathbb{E}(p_b | p_b \geq p_s)}{2} - v_s\right)$$

Using the linear strategy of the buyer we yield

$$\pi_s = \frac{1}{4\delta}(\gamma + \delta - p_s)(3p_s + \gamma + \delta - 4v_s)$$

From FOC we obtain

$$p_s = \frac{2}{3}v_s + \frac{1}{3}(\gamma + \delta)$$

We assumed

$$\begin{aligned} p_b &= \gamma + \delta v_b \\ p_s &= \alpha + \beta v_s \end{aligned}$$

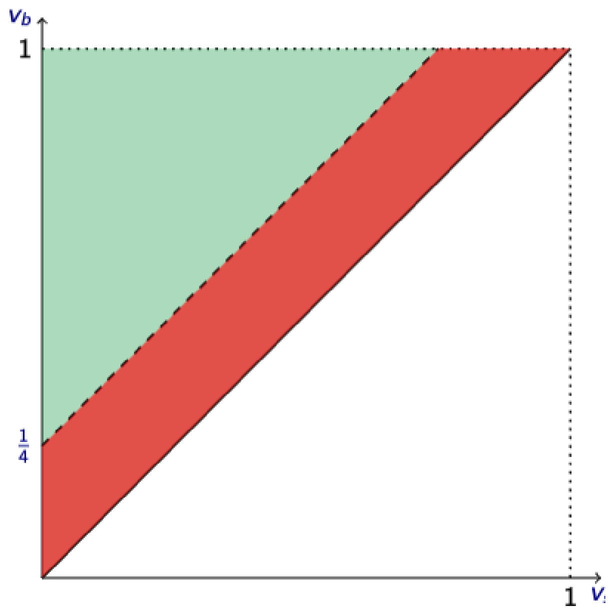
We obtained

$$\begin{aligned} p_b &= \frac{1}{3}\alpha + \frac{2}{3}v_b \\ p_s &= \frac{1}{3}(\gamma + \delta) + \frac{2}{3}v_s \end{aligned}$$

Therefore, the equilibrium strategies are

$$\begin{aligned} p_b &= \frac{1}{12} + \frac{2}{3}v_b \\ p_s &= \frac{1}{4} + \frac{2}{3}v_s \end{aligned}$$

Is this outcome efficient?




The efficient outcome is when trade occurs whenever $v_b \geq v_s$ i.e. in the upper triangle. In equilibrium trade occurs when $p_b \geq p_s$:

$$p_b = \frac{1}{12} + \frac{2}{3} v_b$$
$$p_s = \frac{1}{4} + \frac{2}{3} v_s$$
$$p_b \geq p_s \iff v_b \geq v_s + \frac{1}{4}$$

Thus the outcome is inefficient and the efficiency is reduced by 7/16.
There are other equilibria such as

- Choose some $x \in [0, 1]$
- And we have

$$p_b = \begin{cases} x & \text{if } v_b \geq x \\ 0 & \text{if } v_b < x \end{cases}$$

  and

$$p_s = \begin{cases} x & \text{if } v_s \leq x \\ 1 & \text{if } v_s > x \end{cases}$$

But this turns out to be even less efficient and actually the linear equilibrium is the most efficient one.

**Example 2.9 (Adverse Selection).** There is an asset potentially available for sale. The seller knows the value of the asset under its management. The buyer does not know the value of the asset and thinks its value is uniformly distributed on $[0, 100]$. The buyer believes that it has superior management skills and under its management the value of the asset will go up by 50%. The buyer proposes the price $p$ for the asset, the seller can either accept or reject. If the offer is accepted then the asset changes the ownership.

- Players: buyer and seller
- States: $v \in [0, 100]$
- Actions: Buyer: $p \in \mathbb{R}_+$; seller: $\{a, r\}$
- Signals: the seller gets a perfect signal $s = v$

- Beliefs: Buyer has a uniform belief with support $[0, 100]$
- Payoffs: 0 and $v$ if the offer is rejected, $\frac{3}{2}v - p$ and $p$ if the offer is accepted

The Seller strategy maps prices to $\{a, r\}$. The seller accepts the offer whenever $p \geq v$. Then, the expected payoff of the buyer is

$$\pi_b = \mathbb{P}(p \geq v)\left(\frac{3}{2}\mathbb{E}(v|v \leq p) - p\right) = \frac{p}{100}\left(\frac{3}{2}\frac{p}{2} - p\right) = -\frac{p^2}{400}$$

Thus the buyer's best response to the seller's strategy is to offer $p = 0$. This is an extreme form of adverse selection: any type of asset would be better under the buyer's management.

## 2.3 Harsanyi Purification and Global Games

**Proposition 2.10 (Harsanyi Purification).** The probability distributions over strategies induced by the pure (Bayesian Nash) equilibria of the perturbed game converge to the distribution of the (mixed Nash) equilibrium of the unperturbed game

**Example 2.11.** Consider the typical BoS example

| | $b$ | $s$ |
|---|---|---|
| $B$ | $2,1$ | $0,0$ |
| $S$ | $0,0$ | $1,2$ |

The game has three equilibria: $(B, b), (S, s)$ and $(\frac{2}{3}B + \frac{1}{3}S, \frac{1}{3}b + \frac{2}{3}s)$. Now consider a perturbed game

| | $b$ | $s$ |
|---|---|---|
| $B$ | $2+\varepsilon, 1$ | $\varepsilon, \delta$ |
| $S$ | $0,0$ | $1, 2+\delta$ |

where

- $\varepsilon, \delta$ are players' types (extra payoff from their favourite action), private information of each player
- the prior is $\varepsilon, \delta \sim U[0, x]$

A Bayesian strategy maps the random shock to an action. Suppose that player 2 follows the following pure strategy

- if $\delta > \delta^*$, then play $s$
- if $\delta \leq \delta^*$ then play $b$

What is the best response of player 1? It is optimal to play $B$ whenever:

$$v_1(B) = (2+\varepsilon)\frac{\delta^*}{x} + \varepsilon\left(1 - \frac{\delta^*}{x}\right) \geq 1\left(1 - \frac{\delta^*}{x}\right) = v_1(S) \implies \varepsilon \geq 1 - \frac{3\delta^*}{x}$$

If player 1 plays $B$ whenever $\varepsilon > \varepsilon^*$ then we get that the best response of Player 2 is to play $s$ whenever

$$\delta \geq 1 - \frac{3\varepsilon^*}{x}$$

Intersection of best responses give us

$$\varepsilon^* = \delta^* = \frac{x}{3+x}$$

What is the probability that Player 1 chooses $B$ in this pure strategy BNE?

$$\mathbb{P}(B) = 1 - \frac{\varepsilon^*}{x} = \frac{2+x}{3+x}$$

As we look at vanishingly small uncertainty

$$\lim_{x\to 0}\frac{2+x}{3+x}=\frac{2}{3}$$

So the limit point of outcomes of BNE equilibrium is mixed Nash equilibrium of unperturbed game

**Global Games**: global games is when a game is randomly drawn from a class of games (global games) and each player gets a noisy signal about which game is being played. As types of the players are randomly drawn conditional on the actual game played, types become correlated.

**Example 2.12.** Two investors simultaneously decide on whether they want to participate in a project or not

- participation costs for each investor is $c$
- the project succeeds only when both investors decide to invest
- if the project succeeds it brings $v$ to each investor

| | $i$ | $n$ |
|---|---|---|
| $I$ | $v-c, v-c$ | $-c, 0$ |
| $N$ | $0, -c$ | $0, 0$ |

The problem is that participation costs are unknown ex ante and each player gets a conditionally independent signal about it.
*Complete Information*: Assume that players have complete information about $c$.

- If $c<0$, then the game has a unique NE $(I, i)$
- If $c>v$, then the game has a unique equilibrium $(N, n)$
- If $c\in[0, v]$, then there are three equilibria $(I, i), (N, n)$ and a mixed strategy equilibrium
- Suppose that player 2 invests in a mixed strategy equilibrium with probability $q$, then player 1 is indifferent if
$$\pi_1(I)=q(v-c)-(1-q)c=qv-c=0=\pi_1(N)\implies q=\frac{c}{v}$$
- Note that any player will invest if they think that the probability of another player investing is above $c/v$ and not invest when the expected probability falls below this threshold.

*Correlated Types*: Assume that $v=10$ and $c\in\{-2,-1,\ldots,11,12\}$, all outcomes equally likely. Each player gets a signal about the cost:

- One player learns true value $t=c$
- Another player gets $t\in\{c-1, c+1\}$ both equally likely
- Joint distribution of signals conditional on $c$ is given by

| $t_1/t_2$ | $c-1$ | $c$ | $c+1$ |
|---|---|---|---|
| $c-1$ | 0 | 1/4 | 0 |
| $c$ | 1/4 | 0 | 1/4 |
| $c+1$ | 0 | 1/4 | 0 |

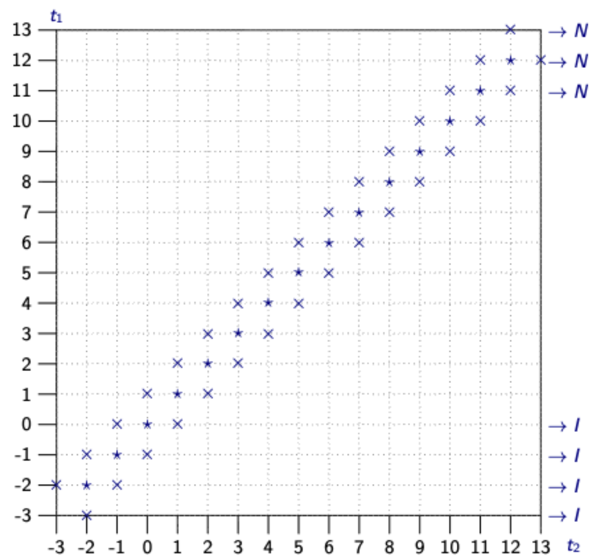
Players have almost perfect knowledge of the cost but they do not have common knowledge e.g. a type $t$ player 1 could think that player 2 thinks that player 2 thinks that player 2 thinks that player 1 has a type $t+4$ even though types must be only one apart.
Suppose that player 1 gets a signal $t_1\in[-1, 11]$. Then, player 1 belief about the costs is

$$c|t_1=\begin{cases}t_1-1 & \text{w.p. } 1/4\\ t_1 & \text{w.p. } 1/2\\ t_1+1 & \text{w.p. } 1/4\end{cases}\implies \mathbb{E}[c|t_1]=t_1$$

If $t_1 = -3$, then player 1 knows that $c = -2$ since the costs can only be from -2 onwards. If $t_1 = -2$ then $c = -2$ w.p. $2/3$ or $c = -1$ w.p. $1/3$ (from Bayes rule). Cases $t = 12, 13$ are symmetric. Belief of player 1 about player 2 signal for $t_1 \in [-1, 11]$.

$$t_1 | t_2 = \begin{cases} t_1 - 1 & \text{w.p. } 1/2 \\ t_1 + 1 & \text{w.p. } 1/2 \end{cases}$$

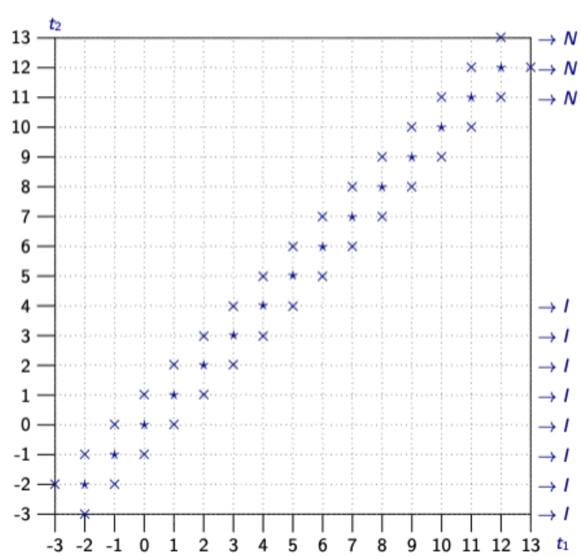
If $t_1 = -3$ then player 1 knows that $t_2 = -2$. If $t_1 = -2$, then $t_2 = -3$ w.p. $1/4$ and $t_2 = -1$ w.p. $3/4$. We now look at IESDS.
Step 1:




Notice that if $t_1 < 0$, then $\mathbb{E}[c|t_1] < 0$ so investing is a dominant action. If $t_1 > 10$, then $\mathbb{E}[c|t_1] > 10$ so not investing is a dominant action. Suppose $t_1 = 0$, then the expected payoff of player 1 choosing $I$ is

$$\mathbb{P}[a_2 = i | t_2 = 0] \times v - \mathbb{E}[c|t_1] \geq \mathbb{P}[t_2 = -1 | t_1 = 0] \times v - 0 = v_2 > 0$$

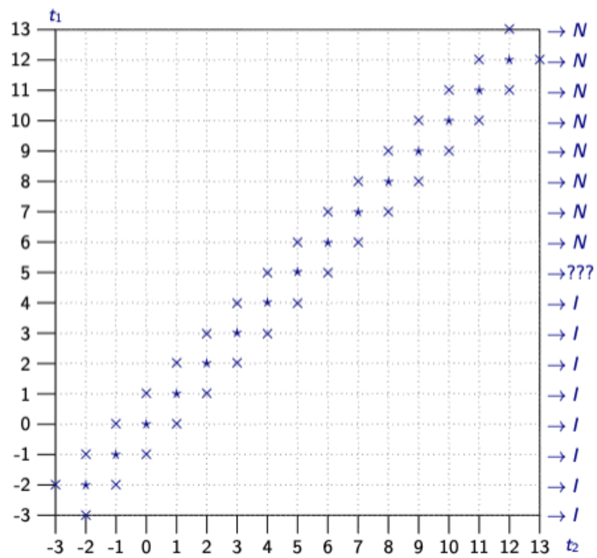
Thus, player 1 should choose $I$.
Step 2:



By continuing the elimination, if both types invest for $t_i = t-1$, then investing at $t_i = t$ is strictly dominant if

$$\mathbb{P}[I|t_1 = t] \times v - \mathbb{E}[c|t_i] \geq \mathbb{P}[t_{-i} = t-1|t_i = t] \times v - t = v/2 - c$$

Thus, for $c < 5$, investing is a dominant strategy.
Step 3:




Suppose $t_1 = 10$ and then the expected payoff of player 1 choosing $I$ is

$$\mathbb{P}[a_2 = i|t_1 = 10] \times v - \mathbb{E}[c|t_1] \leq (1 - \mathbb{P}[t_2 = 11|t_1 = 10]) \times 10 - 10 < 0$$

Thus, player 1 should choose $N$. By continuing the elimination, if both types do not invest for $t_i = t+1$, then investing at $t_i = t$ is strictly dominant if

$$\mathbb{P}[a_{-i} = I|t_i = t] \times v - \mathbb{E}[c|t_i] \leq (1 - \mathbb{P}[t_{-i} = t+1|t_i = t]) \times v - t = v/2 - c$$

Thus for $c > 5$, not investing is a dominant strategy.

Under complete information for $c \in [0, 10]$ each game had three equilibria. Introducing correlated uncertainty delivers a unique prediction for almost all values of $c$. This conclusion is replicated in a game with a continuum of types and holds even if uncertainty is vanishingly small. Thus global games gives a simple tool for equilibrium selection. The outcome obtained corresponds to a situation when a player has a "Laplacian prior" over opponent's action: each action is taken with equal probability

- In this case, a player should invest whenever $v/2 > c$ and abstain from investing if $v/2 < c$ – same as in our BNE

**Definition 2.13 (Risk Dominance).** A risk dominant equilibrium is an equilibrium with higher deviation payoff product

**Example 2.14.** Consider

| | $L$ | $R$ |
|---|---|---|
| $T$ | $a, b$ | $c, d$ |
| $B$ | $e, f$ | $g, h$ |

Suppose that $a > e, b > d, g > c, h > f$. Thus, there are two pure strategy equilibria: $(T, L)$ and $(B, R)$:

- The deviation-payoff product for the equilibrium $(T, L)$ is $(a - e)(b - d)$

- The deviation-payoff product for the equilibrium $(B, R)$ is $(g-c)(h-f)$
- If $(a-e)(b-d) > (g-c)(h-f)$ then $(T, L)$ is risk-dominant. If $(a-e)(b-d) < (g-c)(h-f)$ then $(B, R)$ is risk-dominant. If both deviation payoff products are equal, then neither equilibrium is risk-dominant.

Notice that in a symmetric game, risk dominance coincides with the equilibrium made up of strategies that are (strict) best responses to a 50/50 mix by the opponent.

**Example 2.15.**

| | $L$ | $R$ |
|---|---|---|
| $T$ | $a, a$ | $b, c$ |
| $B$ | $c, b$ | $d, d$ |

$(a-c)^2 > (d-b)^2 \iff a-c > d-b \iff \frac{1}{2}a + \frac{1}{2}b > \frac{1}{2}c + \frac{1}{2}d.$
Recall that the equilibrium in the global game also involved the best response to 50:50 mix:

- Take the general class of 2 by 2 games with pure equilibria
- Suppose that payoffs are initially unknown
- Each player receives a very accurate but private signal of the payoffs
- Unique equilibrium is to play the risk-dominant equilibrium indicated by the signal

**Proposition 2.16.** Risk dominance is not transitive

*Proof.* Consider the following game

| | $L$ | $M$ | $R$ |
|---|---|---|---|
| $T$ | $6, 6$ | $0, 3$ | $2, 0$ |
| $C$ | $3, 0$ | $5, 5$ | $1, 4$ |
| $B$ | $0, 2$ | $4, 1$ | $4, 4$ |

Notice that $(T, L)$ risk dominates $(B, R)$ which risk dominates $(C, M)$ which risk dominates $(T, L)$. ■

The upshot of this is that risk-dominance is a pairwise criterion

# 3 Knowledge in Games

## 3.1 Modelling Common Knowledge

The basis of the model is a set $\Omega$ of states (one interpretation of this is a possible world). An event $E \subseteq \Omega$ is true in state $\omega$ if $\omega \in E$. Note that this implies that an event is a set, not an element.

We want to define the extent of a decision-maker's knowledge which means that we have to specify an information function $P$ that associates every state $\omega \in \Omega$ with a non-empty subset $P(\omega)$ of $\Omega$. The interpretation is that when the state is $\omega$ the decision-maker knows only that the state is in the set $P(\omega)$ i.e. he considers it possible that the true state could be any state in $P(\omega)$ but not in any state outside $P(\omega)$.

**Definition 3.1 (Information Function).** An information function for the set $\Omega$ of states is a function $P$ that associates with every state $\omega \in \Omega$ a nonempty subset $P(\omega)$ of $\Omega$. The pair $\rangle \Omega, P \langle$ consisting of the states and information function satisfy the following conditions:

- P1: $\omega \in P(\omega)$ for every $\omega \in \Omega$
- P2: If $\omega' \in P(\omega)$, then $P(\omega') = P(\omega)$

P1 says that the decision-maker never excludes the true state from the set of states he regards as feasible: he is never certain that the state is different from the true state. P2 says that the decision-maker uses the consistency or inconsistency of states with his information to make inferences about the state.

The following condition is equivalent to P1 and P2

**Definition 3.2.** An information function $P$ for the set $\Omega$ of states is partitional if there is a partition of $\Omega$ such that for any $\omega \in \Omega$, the set $P(\omega)$ is the element of the partition that contains $\omega$

**Proposition 3.3.** An information function is partitional iff it satisfies P1 and P2

*Proof.* If $P$ is partitional then it clearly satisfies P1 and P2. Now suppose that P satisfies P1 and P2. If $P(\omega)$ and $P(\omega')$ intersect and $\omega'' \in P(\omega) \cap P(\omega')$ then by P2 we have $P(\omega) = P(\omega') = P(\omega'')$; by P1, we have $\cup_{\omega \in \Omega} P(\omega) = \Omega$. Thus $P$ is partitional ■

We now can model the knowledge of a decision-maker via the *Knowledge Function.* We refer to a set of states (a subset of $\Omega$) as an event. Given our interpretation of an information function, a decision-maker for whom $P(\omega) \subseteq E$ knows, in the state $\omega$, that some state in the event $E$ has occurred. In this case we say that the state $\omega$ in the decision-maker knows $E$

**Definition 3.4 (Knowledge Function).** Given an information function $P$, the decision-maker's knowledge function $K$ is given by

$$K(E) = \{\omega \in \Omega : P(\omega) \subseteq E\}$$

For any event $E$, the set $K(E)$ is the set of all states in which the decision maker knows $E$. The knowledge function $K$ that is derived from any information function satisfies the following three properties

- K1: $K(\Omega) = \Omega$
- K2: If $E \subseteq F$, then $K(E) \subseteq F$
- K3: $K(E) \cap K(F) = K(E \cap F)$

K1 says that in all states the decision maker knows that some state in $\Omega$ has occurred. K2 says that if $F$ occurs whenever $E$ occurs and the decision maker knows $E$ then he knows $F$: if $E$ implies $F$, then knowledge of $E$ implies knowledge of $F$. K3 means that if the decision maker knows both $E$ and $F$ then he knows $E \cap F$.

**Proposition 3.5 (Further properties of the knowledge function).** The knowledge function has the following properties depending on the conditions that our information function satisfy:

- Suppose $P$ satisfies P1. Then

$$\text{K4 (Axiom of Knowledge)} \quad K(E) \subseteq E$$

- Suppose $P$ satisfies P1 and P2 (and hence is partitional). Then, the knowledge function satisfies two further properties

$$\text{K5 (Axiom of Transparency)} \quad K(E) \subseteq K(K(E))$$

and

$$\text{K6 (Axiom of Wisdom)} \quad \Omega \backslash K(E) \subseteq K(\Omega \backslash K(E))$$

We may also give an interpretation of the information function in terms of a knowledge function (where we define it as a function that associates a subset of $\Omega$ with each event $E \subseteq \Omega$

**Proposition 3.6.** Our information function $P$, taking our knowledge function as primitive, is: for each state $\omega$, let

$$P(\omega) = \cap\{E \subseteq \Omega : \omega \in K(E)\}$$

If there is no event $E$ for which $\omega \in K(E)$ then we take the intersection to be $\Omega$

We can now model common knowledge. The event that everybody knows $E$ is

$$K(E) = \bigcap_i K_i(E) = \{\omega \in \Omega : P_i(\omega) \subseteq E \text{ for all } i\}$$

The event that everybody knowledge that everybody knows $E$ is

$$K^2(E) = K(K(E))$$

More generally,

$$K^n(E) = \{\omega \in \Omega : P_i(\omega) \subseteq K^{n-1}(E) \text{ for all } i\}$$

**Definition 3.7 (Common Knowledge).** The event that $E$ is commonly known is

$$K^{\infty}(E) = \bigcap_n K^n(E)$$

and the event $E$ is common knowledge in state $\omega$ if $\omega \in K^{\infty}(E)$. In other words, an event $E \in \Omega$ is common knowledge in $\omega \in \Omega$ if $\omega$ is a member of every set in the infinite sequence $K_1(E), K_2(E), \ldots, K_1(K_2(E)), K_2(K_1(E)), \ldots$

Alternatively, we have the following definition of common knowledge (which we specify for 2 players)

**Definition 3.8.** Let $P_1$ and $P_2$ be the information functions of individuals 1 and 2 for the set $\Omega$ of states. An event $F \subseteq \Omega$ is self-evident between 1 and 2 if for all $\omega \in F$, we have $P_i(\omega) \subseteq F$ for $i = 1, 2$. An event $E \subseteq \Omega$ is common knowledge between 1 and 2 in the state $\omega \in \Omega$ if there is a self-evident event $F$ for which $\omega \in F \subseteq E$

**Example 3.9.** Let $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6\}$, let $P_1$ and $P_2$ be the partitional information functions of individuals 1 and 2 and let $K_1$ and $K_2$ be the associated knowledge functions. Let the partitions induced by the information functions be

$$\begin{aligned}
\mathcal{P}_1 &= \{\{\omega_1, \omega_2\}, \{\omega_3, \omega_4, \omega_5\}, \{\omega_6\}\} \\
\mathcal{P}_2 &= \{\omega_1\}, \{\omega_2, \omega_3, \omega_4\}, \{\omega_5\}, \{\omega_6\}\}
\end{aligned}$$

The event $E = \{\omega_1, \omega_2, \omega_3, \omega_4\}$ does not contain any event that is self-evident between 1 and 2 and hence in no state is $E$ common knowledge between 1 and 2 in the sense of the second definition. The event $E$ is also not common knowledge in any state in the sense of the first definition since $K_1(K_2(K_1(E))) = \emptyset$ as the following calculations demonstrate

$$\begin{aligned}
K_1(E) &= \{\omega_1, \omega_2\}, \\
K_2(E) &= E, \\
K_2(K_1(E)) &= \{\omega_1\}, \\
K_1(K_2(E)) &= \{\omega_1, \omega_2\}, \\
K_1(K_2(K_1(E))) &= \emptyset, \\
K_2(K_1(K_2(E))) &= \{\omega_1\}
\end{aligned}$$

The event $F = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5\}$ is self-evident between 1 and 2 and hence is common knowledge between 1 and 2 in any state $F$ in the sense of the second definition. Since $K_1(F) = K_2(F) = F$ the event $F$ is also common knowledge between 1 and 2 at any state in $F$ in the sense of the first definition

**Proposition 3.10.** Let $\Omega$ be a finite set of states, let $P_1$ and $P_2$ be the partitional information functions of individuals 1 and 2, and let $K_1$ and $K_2$ be the associated knowledge functions. Then, an event $E \subseteq \Omega$ is common knowledge between 1 and 2 in the state $\omega \in \Omega$ according to **Definition 3.7** iff it is is common knowledge between 1 and 2 in the state $\omega \in \Omega$ according to **Definition 3.8**

### 3.2 "Almost Common Knowledge"

Almost Common Knowledge is to determine whether high order mutual knowledge of an event is "close" to common knowledge of the event.

The motivating example is from Clark and Marshall (1978) Reference Diaries.
Let $R$ denote the intended referent of the definite description (here: the movie "Monkey Business"). Write $K_A\varphi$ for "Ann knows $\varphi$" and $K_B\varphi$ for "Bob knows $\varphi$".

The point: Ann's utterance "What did you think of the movie?" presupposes that $R$ is *sufficiently* mutually known between Ann and Bob. The versions below illustrate increasing depth of higher-order knowledge about $R$.

**Version $\infty$ (Common knowledge benchmark).**

- Scenario: On Wednesday night, Ann and Bob go together to see $R$.
- Knowledge: $R$ is common knowledge between them (all orders):

$$K_A R,\ K_B R,\ K_A K_B R,\ K_B K_A R,\ K_A K_B K_A R,\ \ldots$$

**Version 1 (Only first-order for Ann).**

- Scenario: Ann and Bob both went to see $R$, but neither knows the other went.
- Knowledge:
$$K_A R \quad \text{but Ann does not know } K_B R.$$

**Version 2 (Ann knows Bob knows, but not that Bob knows Ann knows).**

- Scenario: Ann goes to see $R$ and sees Bob there (Bob does not see Ann). Ann realises Bob might have been at other movies too, but she is confident he saw $R$.
- Knowledge:
$$K_A R, \quad K_A K_B R, \quad \text{but Ann does not know } K_B K_A R.$$

**Version 3 (One more level: Ann knows Bob knows that Ann knows).**

- Scenario: Ann sees Bob at $R$; Bob sees Ann (so Bob knows Ann saw $R$), but Ann thinks Bob may not have noticed her.
- Knowledge:

$$K_A R, \quad K_A K_B R, \quad K_A K_B K_A R, \quad \text{but Ann does not know } K_B K_A K_B R.$$

- Interpretation: mutual knowledge holds up to (roughly) 3 levels, but fails at the next.

**Version 4 (Even higher, but still finite).**

- Scenario: Ann sees Bob and Charles at $R$. Ann believes Bob/Charles saw her, but also believes they may think she did not realise they saw her. Later Charles tells Ann that Bob did realise Ann had seen him, but Bob was also sure she did not realise that they had seen her notice them (etc.).
- Knowledge: Ann's higher-order beliefs about $R$ go further than Version 3, but still stop at a finite level:
$$K_A R,\ K_A K_B R,\ K_A K_B K_A R,\ K_A K_B K_A K_B R,\ \ldots$$
for several iterations, but not all.

**Almost common knowledge ("Version $k$").**

- "Version $k$" means: mutual knowledge about $R$ holds up to depth $k$ (i.e. alternating $K_A$ and $K_B$ applied to $R$ up to $k$ times), but may fail at depth $k+1$.
- As $k \to \infty$, Version $k$ approaches the common-knowledge benchmark (Version $\infty$).

## 3.3 No Agreeing to Disagreeing

The main theorem in this section is the "No agreeing to disagreeing" theorem.

**Theorem 3.11 (No Agreeing to Disagreeing).** For $i \in N, E \subseteq \Omega$ and $p \in [0,1]$, let $E_i^p = \{\omega \in \Omega : \mu(E|\mathcal{P}(\omega)) = p\}$ be the event that player $i$ assigns probability $p$ to $E$. Suppose it is common knowledge at $\omega$ that player 1 assigns probability $p$ to $E$ and that player 2 assigns probability $p' = E$. Then $p = p'$

**Interpretation**: Suppose two players begin with identical beliefs about the world (i.e. common prior $\mu$) so that any differences in beliefs is due to differences in information (i.e. $\mathcal{P}_1(\omega)$ and $\mathcal{P}_2(\omega)$). Then, they cannot "agree to disagree" in that they have common knowledge (i.e. agreeing) that they assign different probabilities to the same event under the same information conditions (i.e. disagreeing on how likely).
This is one implication of the "agreement" theorem.

**Example 3.12 (No Trade).** Suppose 2 risk neutral players consider betting on event $A$. If $A$ occurs, then player 2 plays \$1 to player 1; if $A$ does not occur, then player 1 pays \$1 to player 2. Denote the posterior probabilities that player 1 and player 2 assign to $A$ by $p$ and $p'$.
Then, player 1 is willing to accept the bet iff $p \geq 1/2$ while player 2 is willing to accept the bet if $p' \leq 1/2$. If either accepts the bet, then this becomes common knowledge. It is impossible that this is common knowledge and that players are strictly willing to trade.

How can beliefs become common knowledge?
**Beliefs and Common Knowledge (finite state examples)**
Throughout let $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$. Player $A$ knows whether the state is "odd/even":

$$\mathcal{P}_A = \big\{\{\omega_1, \omega_3\},\ \{\omega_2, \omega_4\}\big\}.$$

Player $B$ knows the "color" (one state is distinguished):

$$\mathcal{P}_B = \big\{\{\omega_1, \omega_2, \omega_3\},\ \{\omega_4\}\big\}.$$

Write $\mathcal{P}_i(\omega)$ for player $i$'s information set at $\omega$ and $\mu(\cdot \mid \mathcal{P}_i(\omega))$ for posteriors.
**Beliefs and Common Knowledge**

Assume a common prior $\mu$ given by

$$\mu(\omega_1) = \mu(\omega_4) = \frac{1}{3}, \qquad \mu(\omega_2) = \mu(\omega_3) = \frac{1}{6}.$$

Let the event be

$$E = \{\omega_2, \omega_3\}.$$

Suppose the true state is $\omega = \omega_2$.

- $A$'s posterior:
$$\mu(E \mid \mathcal{P}_A(\omega_2)) = \mu(E \mid \{\omega_2, \omega_4\}) = \frac{\mu(\omega_2)}{\mu(\omega_2) + \mu(\omega_4)} = \frac{\frac{1}{6}}{\frac{1}{6} + \frac{1}{3}} = \frac{1}{3}.$$
  So $A$ announces "my posterior that $E$ holds is 1/3."
- $B$'s posterior:
$$\mu(E \mid \mathcal{P}_B(\omega_2)) = \mu(E \mid \{\omega_1, \omega_2, \omega_3\}) = \frac{\mu(\omega_2) + \mu(\omega_3)}{\mu(\omega_1) + \mu(\omega_2) + \mu(\omega_3)} = \frac{\frac{1}{6} + \frac{1}{6}}{\frac{1}{3} + \frac{1}{6} + \frac{1}{6}} = \frac{1}{2}.$$
  So $B$ announces "my posterior that $E$ holds is 1/2."
- From the two announcements (and common knowledge of the model), $A$ can deduce that the true state must lie in $E$ with probability 1 and announces this. Then $B$ also concludes that $E$ holds.

- If instead the true state were $\omega_3$, the same announcement process leads to common knowledge that the posterior probability of $E$ is 1/3.

**Heterogeneous Priors**

Let the event be

$$E = \{\omega_1, \omega_4\}.$$

Assume heterogeneous priors $\mu_A, \mu_B$:

$$\mu_A(\omega_1) = \mu_A(\omega_2) = \mu_A(\omega_3) = \mu_A(\omega_4) = \frac{1}{4},$$

$$\mu_B(\omega_1) = \mu_B(\omega_4) = \frac{2}{10}, \qquad \mu_B(\omega_2) = \mu_B(\omega_3) = \frac{3}{10}.$$

Then for every $\omega \in \Omega$,

$$\mu_A(E \mid \mathcal{P}_A(\omega)) = \frac{1}{2}, \qquad \mu_B(E \mid \mathcal{P}_B(\omega)) = \frac{2}{5}.$$

**Posterior is not Common Knowledge**

Assume a common prior that is uniform:

$$\mu(\omega_k) = \frac{1}{4} \quad \text{for } k = 1, 2, 3, 4,$$

with the same information partitions $\mathcal{P}_A, \mathcal{P}_B$ as above. Let

$$E = \{\omega_1, \omega_2\}.$$

- $A$'s posterior is 1/2 in every state:

$$\mu(E \mid \mathcal{P}_A(\omega_1)) = \mu(E \mid \{\omega_1, \omega_3\}) = \frac{1}{2}, \qquad \mu(E \mid \mathcal{P}_A(\omega_2)) = \mu(E \mid \{\omega_2, \omega_4\}) = \frac{1}{2}.$$

- $B$'s posterior depends on his information set:

$$\mu(E \mid \mathcal{P}_B(\omega)) = \mu(E \mid \{\omega_1, \omega_2, \omega_3\}) = \frac{2}{3} \quad \text{for } \omega \in \{\omega_1, \omega_2, \omega_3\},$$

$$\mu(E \mid \mathcal{P}_B(\omega_4)) = \mu(E \mid \{\omega_4\}) = 0.$$

- At $\omega_2$, posteriors are mutually known because

$$\mathcal{P}_A(\omega_2) \cap \mathcal{P}_B(\omega_2) = \{\omega_2\}$$

  (i.e. the state is pinned down by combining the two information sets).

- However, there is *no* state in which $B$'s posterior is common knowledge. Indeed, players can have common knowledge that they disagree: $A$'s posterior is always 1/2 whereas $B$'s posterior is either 2/3 or 0.

## 3.4 Epistemology

What are some epistemic conditions for solution concepts?

- Consider a game $\langle N, (A_i)_{i \in N}, (u_i)_{i \in N} \rangle$
- State $\omega \in \Omega$ which specifies for all $i$:
  - The action taken by player $i$, $a_i(\omega)$

  - Player $i$'s beliefs, probability measure $\mu_i(\omega)$ over $A_I{-}i$
  - What $i$ knows: information set $\mathcal{P}_i(\omega)$

- Say that player $i$ is rational in state $\omega$ if he plays a best response given his belief i.e. $a_i(\omega)$ is best response against belief induced by $\mu_i(\omega)$
- Assume that beliefs are consistent with knowledge. That is, the support of $\mu_i(\omega)$ is a subset of $\{a_{-i}(\omega') : \omega' \in \mathcal{P}(\omega)\}$

**Proposition 3.13 (Common Knowledge of rationality implies rationalizability).** Suppose that in state $\omega$ it is common knowledge that all players are rational. Then, for all $i$, $a_i(\omega)$ is rationalizable

**Proposition 3.14.** If $\mu$ is a full support common prior and players are rational in every state, with $a_i(\omega) = a_i(\omega')$ for all $i$ and $\omega, \omega' \in \mathcal{P}_i(\omega)$ then $\langle \Omega, \mu, (\mathcal{P}_i)_i, (a_i)_i \rangle$ is a correlated equilibrium

In other words, if beliefs are derived from a common prior and players are rational in every state, then we obtain a correlated equilibrium.

**Proposition 3.15.** Fix state $\omega$. Then an action profile $a(\omega)$ is a pure Nash Equilibrium if in state $\omega$ for all $i$:

- $i$ knows the other player's actions: $\mathcal{P}_i \subseteq \{\omega' : a_i(\omega') = a_{-i}(\omega)\}$
- $i$ is rational: $a_i(\omega)$ is the best response to $a_{-i}(\omega)$

# 4 Dynamic Games

## 4.1 Complete Information and Subgame Perfect Equilibrium

**Definition 4.1 (Extensive Form Games with Perfect Information).** An extensive form game with perfect information is a tuple $\Gamma_E = \langle N, H, A, P, \succsim \rangle$ where

- Players: There is a set $N$ of players (where nature can be one of the players)
- Histories: There is a set $H$ of sequences that satisfies the following three properties
  1. The empty sequence $\emptyset$ is a member of $H$ (representing the start of the game)
  2. If $(a^k)_{k=1,\dots,K} \in H$ and $L < K$, then $(a^k)_{k=1,\dots,L} \in H$
  3. If an infinite sequence $(a^k)_{k=1}^{\infty}$ satisfies $(a^k)_{k=1,\dots,L} \in H$ for every positive integer $L$ then $(a^k)_{k=1}^{\infty} \in H$

  Each member of $H$ is a history; each component of a history is an action taken by a player. A history $(a^k)_{k=1,\dots,K} \in H$ is terminal if it is infinite or if there is no $a^{K+1}$ such that $(a^k)_{k=1,\dots,K+1} \in H$. The set of terminal histories is denoted $Z$

- Available Actions: For all $h \in H$ define the set of available actions $A(h)$. Terminal histories $Z \subseteq H$ are such that $A(z) = \emptyset$ for all $z \in Z$
- Player Assignments: a function $P$ assigns to each nonterminal history (each member of $H \backslash Z$) a member of $N$ i.e. $P(h) \in N$. If $P(h) =$ Nature, then there is a probability distribution over $A(h)$. Furthermore, $P(z) = \emptyset$ for $z \in Z$
- For each player $i \in N$, there is a preference relation $\succsim_i$ on $Z$

**Remark.** By defining $H$ as a set of sequences, we are implicitly excluding any games with continuous histories i.e. only considering countable histories.

We may represent an extensive form in either a game tree or a reduced strategic (normal) form.

**Example 4.2.** Consider the following extensive-form game. Ann moves first and chooses either $B$ or $S$. Bob then moves, but his two decision nodes are in the same information set, so he does not know whether Ann chose $B$ or $S$.

Bob's pure strategies specify what he does after Ann's $B$ and what he does after Ann's $S$. Hence his strategy set is

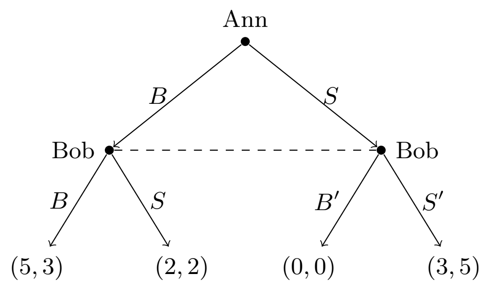
$$\{BB',\, BS',\, SB',\, SS'\},$$

where the first letter denotes Bob's action after Ann chooses $B$, and the second letter denotes Bob's action after Ann chooses $S$.

The strategic-form representation is:

| | $BB'$ | $BS'$ | $SB'$ | $SS'$ |
|---|---|---|---|---|
| $B$ | $(5,3)$ | $(5,3)$ | $(2,2)$ | $(2,2)$ |
| $S$ | $(0,0)$ | $(3,5)$ | $(0,0)$ | $(3,5)$ |

The extensive form is:




Note that Bob, after observing Ann, has **singleton information sets** rather than the dashed line because he knows what Ann has played. The point of the information set, before this observation of what Ann has played, is merely to illustrate that we denote information sets through such a dotted line.

In this example, notice that there are 3 nash equilibria i.e. $(B, BB'), (B, BS'), (S, SS')$ but there is only one subgame perfect equilibria $(B, BS')$ that is sequentially rational. The upshot of this is that reduced strategic forms (i.e. matrix representation) looses information.

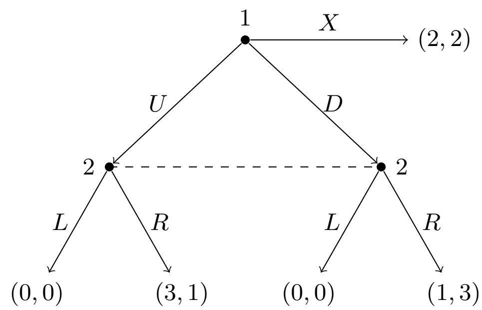
**Definition 4.3 (Subgame).** A subgame of an extensive form game $\Gamma_E$ is a subset of the game having the following properties:

1. It begins with an information set containing a single decision node, contains all the decision nodes that are successors (both immediate and later) of this node, and contains only these nodes
2. If decision node $x$ is in the subgame, then every $x' \in H(x)$ is also, where $H(x)$ is the information set that contains decision node $x$. (That is, there are no "broken" information sets)

The game as a whole is also a subgame, as may be some strict subsets of the game (a proper subgame). Hence, if we specify that Bob observes Ann's action and hence has singleton information sets, the game in **Example 4.2** has three different subgames. But there may be extensive form game with no proper subgame (for instance if Bob does not know which branch he is on after Ann moves).

**Example 4.4.** Consider the following extensive-form game. Player 1 moves first and chooses $U$, $D$, or $X$. If Player 1 chooses $X$, the game ends immediately with payoff $(2, 2)$. If Player 1 chooses $U$ or $D$, then Player 2 moves at an information set that contains both of her decision nodes, choosing either $L$ or $R$.

The extensive form is:




Player 1's strategy set is

$$S_1 = \{U, D, X\}.$$

Since Player 2 has only one information set, her strategy set is

$$S_2 = \{L, R\}.$$

Hence the normal-form representation is

| | $L$ | $R$ |
|---|---|---|
| $U$ | $(0, 0)$ | $(3, 1)$ |
| $D$ | $(0, 0)$ | $(1, 3)$ |
| $X$ | $(2, 2)$ | $(2, 2)$ |

There is no proper subgame, because neither of Player 2's decision nodes is a singleton information set. Therefore the whole game is the only subgame, so every Nash equilibrium is subgame perfect.

**Definition 4.5 (Subgame Perfect Equilibrium).** A profile of strategies $\alpha = (\alpha_1, \ldots, \alpha_N)$ is an $N$-player extensive form game $\Gamma_E$ is a subgame perfect Nash Equilibrium if it induces a Nash Equilibrium in every subgame of $\Gamma_E$

**Theorem 4.6.** Every finite game in which players have perfect recall has a subgame perfect equilibrium

**Remark.** Finite recall is the property that players do not forget their own prior moves (not the same as perfect information)

**Proposition 4.7.** In simultaneous move games, all Nash equilibria are subgame perfect equilibria

*Proof.* Follows trivially since simultaneous move games have no proper subgames ■

**Proposition 4.8.** Every finite game of perfect information $\Gamma_E$ has a pure strategy subgame perfect Nash equilibrium. Moreover, if no player has the same payoffs at any two terminal nodes, then there is a unique subgame perfect Nash equilibrium

**Generalised Backward induction**:

1. Start at the end of the game tree, and identify the Nash equilibria for each of the final subgames (i.e. those that have no other subgames needed within them).

2. Select one Nash equilibrium in each of these final subgames and derive the reduced extensive form game in which these final subgames are replaced by the payoffs that result in these subgames when players use these equilibrium strategies

3. Repeat steps 1 and 2 for the reduced game. Continue the procedure until every move in $\Gamma_E$ is determined. This collection of moves at the various information sets of $\Gamma_E$ constitutes a profile of SPNE strategies

4. If multiple equilibria are never encountered at any step of the process, this profile of strategies is the unique SPNE. If multiple equilibria are encountered, the full set of SPNE is identified by repeating the procedure for each possible equilibrium that could occur for the subgames in question

**Example 4.9 (Firm Competition PSet4 Q2).** Two firms $i = 1, 2$ compete in prices. The demand for each firm $i$ is given by

$$D_i = 1 + p_j - p_i,$$

where firm $i$ sets a price $p_i \geq 0$. Note that an increase in firm $j$'s price increases firm $i$'s demand. Suppose there are no costs of production for either firm.
First, suppose that firms set prices simultaneously.

(a) Firm $i$'s profit is

$$\Pi_i(p_i, p_j) = (1 + p_j - p_i)p_i.$$

Taking the derivative with respect to $p_i$,

$$\frac{\partial \Pi_i}{\partial p_i} = 1 + p_j - p_i - p_i = 1 + p_j - 2p_i.$$

Setting this equal to zero gives the best reply

$$p_i = \frac{1 + p_j}{2}.$$

By symmetry,

$$p_j = \frac{1 + p_i}{2}.$$

Substituting,

$$2p_i = 1 + \frac{1 + p_i}{2}$$

so

$$4p_i = 2 + 1 + p_i,$$

hence

$$3p_i = 3 \iff p_i = 1.$$

Therefore also $p_j = 1$, so the Nash equilibrium is

$$(p_1^*, p_2^*) = (1, 1).$$

Equilibrium profits are

$$(\Pi_1, \Pi_2) = (1, 1).$$

The best-reply functions are upward sloping, reflecting that the two firms' prices are strategic complements.

(b) Now suppose firm 1 moves first. If firm 1 chooses $p_1$, then firm 2's profit is

$$\Pi_2(p_1, p_2) = (1 + p_1 - p_2)p_2.$$

Taking the derivative with respect to $p_2$,

$$\frac{\partial \Pi_2}{\partial p_2} = 1 + p_1 - p_2 - p_2 = 1 + p_1 - 2p_2.$$

Setting this equal to zero gives

$$p_2 = \frac{1 + p_1}{2}.$$

Substituting this continuation best reply into firm 1's objective,

$$\Pi_1\big(p_1, p_2(p_1)\big) = \big(1 + p_2(p_1) - p_1\big)p_1 = \left(1 + \frac{1+p_1}{2} - p_1\right)p_1 = (1.5 - 0.5p_1)\,p_1.$$

Differentiate with respect to $p_1$:

$$\frac{d\Pi_1}{dp_1} = 1.5 - 0.5p_1 - 0.5p_1 = 1.5 - p_1.$$

Setting this equal to zero,

$$p_1 = 1.5.$$

Therefore

$$p_2 = \frac{1+1.5}{2} = 1.25.$$

So the SPE is

$$(p_1^*, p_2^*) = (1.5, 1.25).$$

The induced profits are

$$\Pi_1 = (1 + 1.25 - 1.5)(1.5) = 1.125,$$

and

$$\Pi_2 = (1 + 1.5 - 1.25)(1.25) = 1.5625.$$

Thus

$$(\Pi_1, \Pi_2) = (1.125,\ 1.5625).$$

This SPE is unique, since firm 2 has a unique best reply

$$p_2(p_1) = \frac{1+p_1}{2}$$

for every $p_1$, and firm 1's problem

$$\Pi_1(p_1) = (1.5 - 0.5p_1)\,p_1$$

is strictly concave.

(c) In the simultaneous-move game,

$$(\Pi_1, \Pi_2) = (1, 1),$$

while in the sequential game,

$$(\Pi_1, \Pi_2) = (1.125,\ 1.5625).$$

Hence the sequential game is a Pareto improvement relative to the simultaneous game, since both firms earn higher profits.

However, there is *no* first-mover advantage here, since

$$1.125 < 1.5625.$$

The follower earns more than the leader. The reason is that prices are strategic complements: a higher price by the leader induces the follower to charge a higher price as well, and the follower can exploit this by adjusting optimally after observing the leader's choice.

(d) Now consider the quantity-choice game. The price is

$$p = \max\{1 - q_1 - q_2, 0\},$$

and firm $i$'s profit is

$$\Pi_i(q_i, q_j) = pq_i.$$

In the simultaneous-move Cournot game, assuming interior solutions,

$$\Pi_i(q_i, q_j) = q_i(1 - q_i - q_j).$$

Maximising with respect to $q_i$,

$$\frac{\partial \Pi_i}{\partial q_i} = 1 - 2q_i - q_j = 0 \iff q_i = \frac{1 - q_j}{2}.$$

Thus the best replies are

$$BR_i(q_j) = \max\left\{\frac{1 - q_j}{2}, 0\right\}, \qquad BR_j(q_i) = \max\left\{\frac{1 - q_i}{2}, 0\right\}.$$

Solving the interior system,

$$q_i = \frac{1 - q_j}{2}, \qquad q_j = \frac{1 - q_i}{2}.$$

Hence

$$2q_i = 1 - q_j.$$

Using $q_i = q_j$ by symmetry,

$$2q_i = 1 - q_i \iff 3q_i = 1 \iff q_i = \frac{1}{3}.$$

Therefore

$$(q_1, q_2) = \left(\frac{1}{3}, \frac{1}{3}\right).$$

There is no corner solution: for example, if $q_i = 0$, then

$$q_j = \frac{1}{2}$$

by the best reply, but then $q_i = 0$ is not a best reply, so no corner can be a Nash equilibrium. Thus the unique Cournot equilibrium is

$$(q_1, q_2) = \left(\frac{1}{3}, \frac{1}{3}\right),$$

with price

$$p = 1 - \frac{1}{3} - \frac{1}{3} = \frac{1}{3},$$

and profits

$$(\Pi_1, \Pi_2) = \left(\frac{1}{9}, \frac{1}{9}\right).$$

In the sequential Stackelberg game, suppose firm 1 chooses $q_1$ first. Firm 2 then solves

$$\max_{q_2} \ \Pi_2(q_1, q_2) = q_2(1 - q_1 - q_2).$$

The first-order condition is

$$\frac{\partial \Pi_2}{\partial q_2} = 1 - q_1 - 2q_2 = 0,$$

so

$$q_2(q_1) = \max\left\{\frac{1 - q_1}{2}, 0\right\}.$$

For firm 1, substituting this into price gives

$$p = 1 - q_1 - q_2 = 1 - q_1 - \frac{1-q_1}{2} = \frac{1-q_1}{2}.$$

Hence firm 1's profit is

$$\Pi_1(q_1) = \frac{1-q_1}{2}q_1 = \frac{q_1 - q_1^2}{2}.$$

Differentiating,

$$\frac{d\Pi_1}{dq_1} = \frac{1}{2} - q_1 = 0,$$

so

$$q_1 = \frac{1}{2}.$$

Then

$$q_2 = \frac{1-\frac{1}{2}}{2} = \frac{1}{4}.$$

Therefore the unique SPE is

$$(q_1^*, q_2^*) = \left(\frac{1}{2}, \frac{1}{4}\right).$$

The profits are

$$\Pi_1 = \left(\frac{1}{4}\right)\left(\frac{1}{2}\right) = \frac{1}{8}, \qquad \Pi_2 = \left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \frac{1}{16}.$$

There is no corner solution: if $q_2 = 0$, then firm 1 would solve

$$\max_{q_1} \; q_1(1-q_1),$$

giving $q_1 = \frac{1}{2}$, but then

$$\Pi_2\left(\frac{1}{2}, 0\right) = 0 < \Pi_2\left(\frac{1}{2}, \frac{1}{4}\right) = \frac{1}{16},$$

so $q_2 = 0$ is not a credible continuation action.

Comparing simultaneous and sequential quantity competition:

$$\text{Cournot profits: } \left(\frac{1}{9}, \frac{1}{9}\right), \qquad \text{Stackelberg profits: } \left(\frac{1}{8}, \frac{1}{16}\right).$$

Hence there *is* a first-mover advantage in the quantity-choice game, since

$$\frac{1}{8} > \frac{1}{16}.$$

The reason is that quantities are strategic substitutes: when the leader produces more, the follower's best response is to produce less. So commitment is valuable in the Stackelberg quantity game, unlike in the price game.

**Example 4.10 (PSet4 Q1).** This question is just pure tedium.

## 4.2 Incomplete Information and Perfect Bayesian Equilibrium

From **Example 4.4**, there is no proper subgame and hence we may not use backward induction as originally defined. Furthermore, we have both $\{(X, pL+(1-p)R) : p \geq \frac{1}{3}\}, (U, R)$ as NE identified in the reduced form. If P1 plays $X$ then P2's move does not matter so he may play $L$. But at P2's information set, no matter what he thinks P1 has done to get there, playing $L$ yields less than playing $R$ and hence is not rational. This is why we need to include beliefs and refine our solution concept of subgame perfection.

**Definition 4.11.** An extensive game is a tuple $\langle N, H, P, f_c, (\mathcal{I}_i)_{i\in N}, (\succsim_i)_{i\in N}\rangle$ that has the following components

- Players: There is a set $N$ of players (where nature can be one of the players)
- Histories: There is a set $H$ of sequences that satisfies the following three properties
  1. The empty sequence $\emptyset$ is a member of $H$ (representing the start of the game)
  2. If $(a^k)_{k=1,\ldots,K} \in H$ and $L < K$, then $(a^k)_{k=1,\ldots,L} \in H$
  3. If an infinite sequence $(a^k)_{k=1}^{\infty}$ satisfies $(a^k)_{k=1,\ldots,L} \in H$ for every positive integer $L$ then $(a^k)_{k=1}^{\infty} \in H$

  Each member of $H$ is a history; each component of a history is an action taken by a player. A history $(a^k)_{k=1,\ldots,K} \in H$ is terminal if it is infinite or if there is no $a^{K+1}$ such that $(a^k)_{k=1,\ldots,K+1} \in H$. The set of terminal histories is denoted $Z$
- Available Actions: For all $h \in H$ define the set of available actions $A(h)$ i.e. $A(h) = \{a : (h, a) \in H\}$. Terminal histories $Z \subseteq H$ are such that $A(z) = \emptyset$ for all $z \in Z$
- Player Assignments: a function $P$ assigns to each nonterminal history (each member of $H\backslash Z$) a member of $N \cup \{c\}$ i.e. $P(h) \in N$. If $P(h) = c$ (nature), then there is a probability distribution over $A(h)$. Furthermore, $P(z) = \emptyset$ for $z \in Z$
- A function $f_c$ that associates every history $h$ for which $P(h) = c$ a probability measure $f_c(\cdot|h)$ on $A(h)$ where such probability measure is independent of every other such measure
- For each player $i \in N$, a partition $\mathcal{I}_i$ of $\{h \in H : P(h) = i\}$ with the property that $A(h) = A(h')$ whenever $h$ and $h'$ are in the same member of the partition. For $I_i \in \mathcal{I}$ we denote by $A(I_i)$ te set $A(h)$ and by $P(I_i)$ the player $P(h)$ for any $h \in I_i$. ($\mathcal{I}_i$ is the information partition of player $i$, a set $I_i \in \mathcal{I}_i$ is an information set of player $i$).
- For each player $i \in N$, there is a preference relation $\succsim_i$ on lotteries over $Z$, that can be represented as the expected value of a payoff function defined on $Z$

But we want to specify not only the players' strategies but also their beliefs at each information set about the history that occurred.

**Definition 4.12 (Assessment).** An assessment in an extensive game is a pair $(\sigma, \mu)$ where $\sigma$ is a profile of strategies and $\mu$ is a function that assigns to every information set a probability measure on the set of histories in the information set.

The sort of games we're interested in are only about the **types** of players – and so the beliefs $\mu$ we specify are about these corresponding types of players. We can define that formally below

**Definition 4.13 (Bayesian extensive game with observable actions).** A Bayesian extensive game with observable actions is a tuple $\langle \Gamma, (\Theta_i), (p_i), (u_i)\rangle$ where

- $\Gamma = \langle N, H, P\rangle$ is an extensive game form with perfect information and simultaneous moves

and for each player $i \in N$

- $\Theta_i$ is a finite set (the set of possible types of player $i$); we write $\Theta = \times_{i\in N}\Theta_i$

- $p_i$ is the probability measure $\Theta_i$ for which $p_i(\theta_i) > 0$ for all $\theta_i \in \Theta_i$ and the measures $p_i$ are stochastically independent ($p_i(\theta_i)$ is the probability that player $i$ is selected to be of tpye $\theta_i$)
- $u_i : \Theta \times Z \to \mathbb{R}$ is a von-Neumann-Morgenstern utility function where $u_i(\theta_i, h)$ is player $i$'s payoff when the profile of types is $\theta$ and the terminal history of $\Gamma$ is $h$

**Definition 4.14 (Perfect Bayesian Equilibrium).** Let $\langle \Gamma, (\Theta_i), (p_i), (u_i)\rangle$ be a Bayesian extensive game with observable actions, where $\Gamma = \langle N, H, P\rangle$. A pair $((\sigma_i), (\mu_i)) = ((\sigma_i(\theta_i))_{i\in N, \theta_i \in \Theta_i}, (\mu_i(h))_{i \in N, h \in H\setminus Z})$ where $\sigma_i(\theta_i)$ is a strategy profile of player $i$ in $\Gamma$ and $\mu_i(h)$ is a probability measure on $\Theta_i$ is a **perfect Bayesian equilibrium** if the following conditions are satisfied:

1. *Sequential Rationality*: For every non-terminal history $h \in H\setminus Z$, every player $i \in P(h)$ and every $\theta_i \in \Theta_i$, the probability measure $O(\sigma_{-i}, \sigma_i(\theta_i), \mu_{-i}|h)$ is the set at least as good for type $\theta_i$ as $O(\sigma_{-i}, s_i, \mu_{-i}|h)$ for any strategy $s_i$ of player $i$ in $\Gamma$
2. *Correct Initial Beliefs*: $\mu_i(\emptyset) = p_i$ for all $i \in N$
3. Action-determined beliefs: If $i \notin P(h)$ and $a \in A(h)$, then $\mu_i(h, a) = \mu_i(h)$; if $i \in P(h), a \in A(h), a' \in A(h)$ and $a_i = a_i'$ then $\mu_i(h, a) = \mu(h, a_i)$
4. *Bayesian Updating*: if $i \in P(h)$ and $a_i$ is in the support of $\sigma_i(\theta_i)(h)$ for some $\theta_i$ in the support of $\mu_i(h)$ then for any $\theta_i' \in \Theta_i$, we have

$$\mu_i(h, a)(\theta_i') = \frac{\sigma_i(\theta_i')(h)(a_i) \cdot \mu_i(h)(\theta_i')}{\sum_{\theta_i \in \Theta_i} \sigma_i(\theta_i)(h)(a_i) \cdot \mu_i(h)(\theta_i)}$$

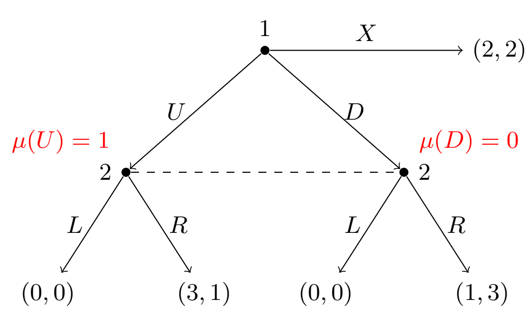
The interpretations of the conditions are as follows:

- Sequential Rationality: Every player chooses optimally given her beliefs at each information set and the others' equilibrium strategies (no restriction on what beliefs they can have about others' types yet)
- Correct Initial Beliefs: at the beginning of the game, there is a commonly known prior which the beliefs must follow
- Action-determined beliefs: only a player $i$'s actions influence the other players' beliefs about his type, which mean (i) if player $i$ does not have a move at $h$ then the actions taken (by others) at $h$ do not affect the beliefs of others about player $i$'s type and (ii) if player $i$ is one of the players who takes an action at $h$ then the other players' beliefs about player $i$'s type depends only on $h$ and the action excludes the possibility that, for instance, player $j$'s updating of his belief about player $i$ is affected by a move made by some player $k \neq i$
- Bayesian Updating: Beliefs must be derived via Bayes rule

**Remark.** It is not necessarily the case that PBE can only be about types but can also be about beliefs over which nodes will be reached. In general the principle to find a PBE is just

1. Sequential Rationality: Every player chooses optimally given her beliefs at each information set and the others' equilibrium strategies
2. Bayesian Beliefs

**Example 4.15.** Consider the following extensive-form game:



The strategic form is

| | $L$ | $R$ |
|---|---|---|
| $U$ | $(0,0)$ | $(3,1)$ |
| $D$ | $(0,0)$ | $(1,3)$ |
| $X$ | $(2,2)$ | $(2,2)$ |

where Player 1 chooses from $\{U, D, X\}$ and Player 2 chooses from $\{L, R\}$.

Let $\mu$ denote Player 2's belief at his information set that he is at the *left* node (that is, that Player 1 chose $U$). Then $1-\mu$ is the probability of being at the right node (that is, that Player 1 chose $D$).

We claim that the unique perfect Bayesian equilibrium is

$$(U, R; \mu = 1).$$

**Why is this a PBE?**

A perfect Bayesian equilibrium requires:

1. sequential rationality of actions given beliefs;
2. consistency of beliefs with equilibrium play via Bayes' rule whenever possible.

*Step 1: Player 2's optimal action given belief $\mu$.* If Player 2 chooses $L$, his payoff is always 0, regardless of which node in the information set has been reached. Hence

$$u_2(L \mid \mu) = 0.$$

If Player 2 chooses $R$, then his payoff is 1 at the left node and 3 at the right node, so

$$u_2(R \mid \mu) = \mu \cdot 1 + (1-\mu) \cdot 3 = 3 - 2\mu.$$

Since $0 \le \mu \le 1$, we have

$$3 - 2\mu \ge 1 > 0.$$

Therefore

$$u_2(R \mid \mu) > u_2(L \mid \mu) \quad \text{for every } \mu \in [0,1].$$

So $R$ is the unique sequentially rational action for Player 2, no matter what his belief is.

*Step 2: Player 1's optimal action anticipating $R$.* Since Player 2 must play $R$, Player 1 compares the payoffs:

$$U \mapsto 3, \qquad D \mapsto 1, \qquad X \mapsto 2.$$

Hence Player 1's unique best response is $U$.

*Step 3: Consistency of beliefs.* Since Player 1 chooses $U$ with probability 1 in equilibrium, Bayes' rule implies that at Player 2's information set,

$$\mu = 1.$$

Thus the assessment $(U, R; \mu = 1)$ is consistent and sequentially rational, so it is a PBE.

**Why is it unique?**

Suppose $(s_1, s_2; \mu)$ is any PBE. Because $R$ yields Player 2 a strictly higher payoff than $L$ for every belief $\mu \in [0,1]$, sequential rationality forces

$$s_2 = R.$$

Given that Player 2 plays $R$, Player 1 strictly prefers

$$U \succ X \succ D$$

since his payoffs are $3, 2, 1$, respectively. So sequential rationality forces

$$s_1 = U.$$

Finally, because $U$ is played with probability 1, Bayes' rule implies

$$\mu = 1.$$

Hence every PBE must be exactly

$$(U, R; \mu = 1).$$

Therefore the PBE is unique. (Note: because this does not include types, only $\mu = 1$ is possible as player 1 must play $U$)

**Remark.** This example illustrates why PBE is more demanding than Nash equilibrium in imperfect-information games. The Nash equilibrium $(X, L)$ is not a PBE, because $L$ is not sequentially rational at Player 2's information set: regardless of his belief, Player 2 strictly prefers $R$.

**Theorem 4.16 (Tirole and Fudenberg, 1991).** Every finite extensive form game with perfect recall has a PBE

**Definition 4.17 (Sequentially Rational Assessments).** Let $\Gamma = \langle N, H, P, f_c, (\mathcal{I}_i), (\succsim_i)\rangle$ be an extensive game with perfect recall. The assessment $(\sigma, \mu)$ is sequentially rational if for every player $i \in N$, and every information set $I_i \in \mathcal{I}_i$ we have

$$O(\sigma, \mu|I_i) \succsim_i O((\sigma_{-i}, \sigma_i'), \mu|I_i) \text{ for every strategy } \sigma_i' \text{ of player } i$$

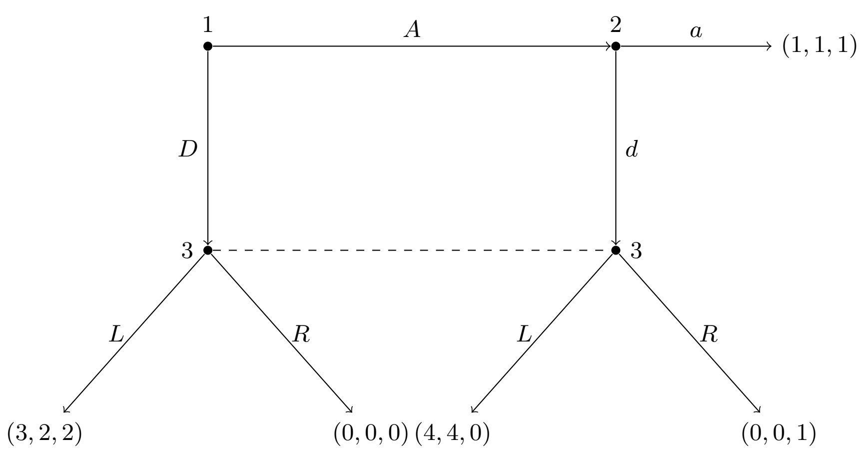
**Definition 4.18 (Consistency).** Let $\Gamma = \langle N, H, P, f_c, (\mathcal{I}_i), (\succsim_i)\rangle$ be a finite extensive game with perfect recall. An assessment $(\sigma, \mu)$ is consistent if there is a sequence $((\sigma^n, \mu^n))_{n=1}^{\infty}$ of assessments that converges to $(\sigma, \mu)$ in Euclidean space and has the properties that each strategy profile $\sigma^n$ is completely mixed and that each belief system $\mu^n$ is derived from $\sigma^n$ using Bayes rule

**Definition 4.19 (Sequential Equilibrium).** An assessment $(\sigma, \mu)$ is a sequential equilibrium of a finite extensive game with perfect recall if it is sequentially rational and consistent

**Remark.** Basically, this says that off-path beliefs must be the limit of Bayesian beliefs generated by fully-mixed strategies ('trembles'); otherwise same as PBE

**Theorem 4.20 (Kreps & Wilson, 1982).** Every finite game with perfect recall has a sequential equilibrium. All Sequential Equilibrium are Perfect Bayesian Equilibrium, and all Perfect Bayesian Equilibrium are Subgame Perfect Equilibrium

**Example 4.21 (Selten's horse).** Consider the following extensive-form game:




Player 1 moves first and chooses $A$ or $D$. If player 1 chooses $A$, then player 2 chooses $a$ or $d$. If player 1 chooses $D$, or if player 1 chooses $A$ and player 2 chooses $d$, then player 3 moves at a single information set and chooses $L$ or $R$.
The pure strategy sets are

$$S_1 = \{A, D\}, \qquad S_2 = \{a, d\}, \qquad S_3 = \{L, R\}.$$

The normal form can be written as two matrices, one for each action of player 3.

If player 3 plays $L$, the payoff matrix is

| | $a$ | $d$ |
|---|---|---|
| $A$ | $(1, 1, 1)$ | $(4, 4, 0)$ |
| $D$ | $(3, 2, 2)$ | $(3, 2, 2)$ |

If player 3 plays $R$, the payoff matrix is

| | $a$ | $d$ |
|---|---|---|
| $A$ | $(1, 1, 1)$ | $(0, 0, 1)$ |
| $D$ | $(0, 0, 0)$ | $(0, 0, 0)$ |

**Find Nash Equilibrium**:
Let player 3's mixed strategy be $qL + (1 - q)R$, where $q \in [0, 1]$.
Given $q$, player 2's payoff from $a$ after $A$ is always

$$u_2(a) = 1,$$

whereas the payoff from $d$ is

$$u_2(d) = 4q.$$

Hence player 2 prefers

$$a \iff 1 \geq 4q \iff q \leq \tfrac{1}{4},$$
$$d \iff 4q \geq 1 \iff q \geq \tfrac{1}{4},$$

and is indifferent when $q = \frac{1}{4}$.
Next, if player 2 plays $a$, then player 1's payoff is

$$u_1(A) = 1, \qquad u_1(D) = 3q.$$

So player 1 prefers

$$A \iff 1 \geq 3q \iff q \leq \tfrac{1}{3},$$
$$D \iff 3q \geq 1 \iff q \geq \tfrac{1}{3},$$

and is indifferent when $q = \frac{1}{3}$.
If instead player 2 plays $d$, then player 1's payoff is

$$u_1(A) = 4q, \qquad u_1(D) = 3q,$$

so for any $q > 0$, player 1 strictly prefers $A$, and if $q = 0$ then player 1 is indifferent between $A$ and $D$.
Finally, given $(A, a)$, player 3 is off path and so any action is compatible with Nash equilibrium. Given $(D, a)$, player 3 gets

$$u_3(L) = 2, \qquad u_3(R) = 0,$$

so $L$ is optimal. Given $(A, d)$, player 3 gets

$$u_3(L) = 0, \qquad u_3(R) = 1,$$

so $R$ is optimal.
Therefore the pure Nash equilibria are

$$(D, a, L) \qquad \text{and} \qquad (A, a, R).$$

There are also mixed Nash equilibria. Since player 2 is willing to play $a$ whenever $q \leq \frac{1}{4}$, and player 1 is willing to play $D$ whenever $q \geq \frac{1}{3}$, there is no interior mixed equilibrium with both $A$ and $D$ used when player 2 plays $a$ purely. Instead, the mixed Nash families are:

$$\{(D,\ pa + (1 - p)d,\ L) :\ p \geq \tfrac{1}{3}\},$$

and

$$\{(A,\ a,\ qL + (1-q)R):\ q \le \tfrac{1}{4}\}.$$

To see the first family, if player 3 plays $L$, then player 2's payoffs from $a$ and $d$ are both 2 when player 1 plays $D$, so player 2 may mix arbitrarily. If player 2 plays $a$ with probability $p$, then player 1's payoff from $A$ is $p$, while from $D$ it is always 3. Thus $D$ is optimal for any $p$, and player 3 prefers $L$ whenever $D$ is reached. To see the second family, if player 1 plays $A$ and player 2 plays $a$, then player 3 is off path and may mix. If player 3 uses $qL + (1-q)R$, then player 2 prefers $a$ whenever $q \le \frac{1}{4}$, and player 1 prefers $A$ since $1 \ge 3q$ for all $q \le \frac{1}{4}$.
We now derive the perfect Bayesian equilibria.
Let

$$\mu_D = \Pr(\text{player 3 is at the left node } \mid \text{player 3 is called to move}),$$

that is, player 3's belief that player 1 chose $D$. Then $1 - \mu_D$ is the probability that the right node was reached, i.e. the history is $(A, d)$.
At player 3's information set, if she chooses $L$, her expected payoff is

$$U_3(L) = 2\mu_D.$$

If she chooses $R$, her expected payoff is

$$U_3(R) = 1 - \mu_D.$$

Hence

$$L \text{ is optimal} \iff 2\mu_D \ge 1 - \mu_D \iff \mu_D \ge \tfrac{1}{3},$$

and

$$R \text{ is optimal} \iff 1 - \mu_D \ge 2\mu_D \iff \mu_D \le \tfrac{1}{3}.$$

**Show that $(D, a, L)$ is not part of PBE**:
We first show that $(D, a, L)$ is not part of any PBE.
Suppose towards contradiction that some PBE had outcome $(D, a, L)$. Then player 3 must choose $L$, so sequential rationality at player 3's information set requires

$$\mu_D \ge \tfrac{1}{3}.$$

But if player 2 were reached, then choosing $a$ gives payoff 1, while choosing $d$ leads to player 3, who is supposed to play $L$, giving player 2 payoff 4. Thus $d$ is strictly better than $a$, so $a$ is not sequentially rational. Therefore $(D, a, L)$ cannot be a PBE.
**Show that $(A, a, R)$ is PBE**:
Now consider $(A, a, R)$. If player 3 holds a belief $\mu_D \le \frac{1}{3}$, then $R$ is sequentially rational. Given that player 3 plays $R$, if player 2 is reached she compares

$$a \mapsto 1, \qquad d \mapsto 0,$$

so $a$ is sequentially rational. Given $(a, R)$, player 1 compares

$$A \mapsto 1, \qquad D \mapsto 0,$$

so $A$ is sequentially rational. Therefore

$$(A, a, R; \mu_D)$$

is a PBE for every $\mu_D \le \frac{1}{3}$.
At the boundary $\mu_D = \frac{1}{3}$, player 3 is indifferent between $L$ and $R$, since

$$2\mu_D = 1 - \mu_D = \tfrac{2}{3}.$$

So player 3 may mix, choosing $qL + (1-q)R$. For player 2 still to prefer $a$, we need

$$1 \ge 4q \iff q \le \tfrac{1}{4}.$$

Thus there are also PBE assessments of the form

$$(A, a,\ qL + (1-q)R;\ \mu_D = \tfrac{1}{3}), \qquad q \le \tfrac{1}{4}.$$

In conclusion, the Nash equilibria are

$$(D, a, L), \qquad (A, a, R),$$

together with the mixed families

$$\{(D,\ pa + (1-p)d,\ L) : p \ge \tfrac{1}{3}\}$$

and

$$\{(A,\ a,\ qL + (1-q)R) : q \le \tfrac{1}{4}\}.$$

However, the only PBE outcomes are those with path

$$A \to a \to (1,1,1).$$

So PBE eliminates the equilibrium outcome based on $(D, a, L)$ and selects the unique outcome

$$(1,1,1).$$

**Remark.** Notice that we proceed by looking at all the NE in the reduced normal form and then try to see if these are PBE as well under certain beliefs. This is because the PBE is a subset of the NE

**Example 4.22 (Centipede with doubt).** Consider a centipede game in which Player 1 may be one of two types:

crazy type $c$ or rational type $r$.

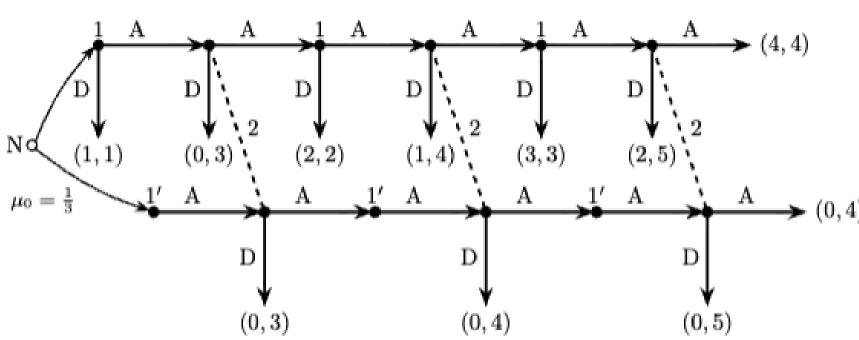
The crazy type always plays $A$. The rational type chooses actions strategically. At the start of the game, Nature selects Player 1's type. Player 2 assigns prior probability

$$\mu_0 = \frac{1}{3}$$

to the event that Player 1 is crazy, and probability

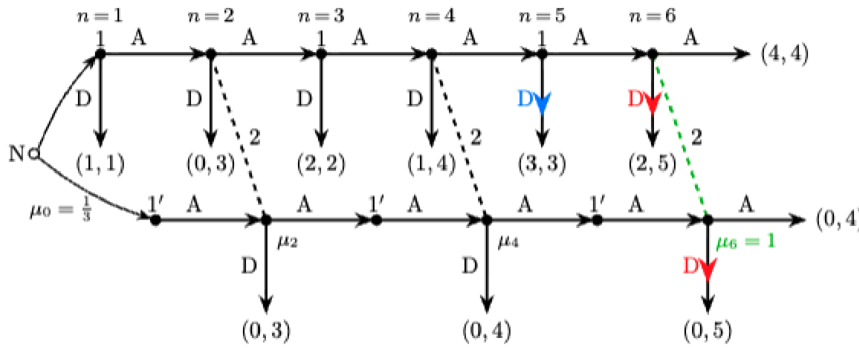
$$1 - \mu_0 = \frac{2}{3}$$

to the event that Player 1 is rational.
The game tree is:




Let $\mu_2, \mu_4, \mu_6$ denote Player 2's posterior probability that Player 1 is crazy at her three information sets.

**Step 1: Verify the endgame.**




At $n = 6$, Player 2 compares

$$D \mapsto 5, \qquad A \mapsto 4.$$

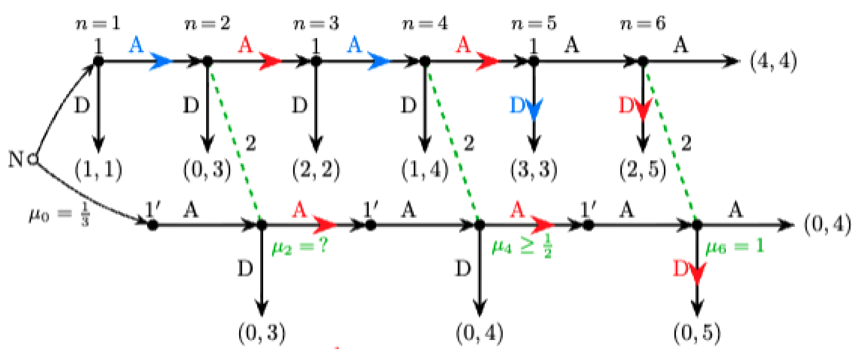
Hence Player 2 strictly prefers $D$ at $n = 6$, regardless of beliefs.
At $n = 5$, the rational type of Player 1 then compares

$$D \mapsto 3, \qquad A \mapsto 2,$$

so the rational type strictly prefers $D$.
Therefore, if Player 2 reaches her last information set and observes that Player 1 continued at $n = 5$, she infers that Player 1 must be crazy. Hence

$$\mu_6 = 1.$$

**Step 2: Verify that Player 2 cannot play a pure action at $n = 4$.**




At $n = 4$, if Player 2 chooses $D$, her payoff is

$$4.$$

If she chooses $A$, her expected payoff is

$$5\mu_4 + 3(1 - \mu_4) = 3 + 2\mu_4.$$

Hence:

$$D \text{ is optimal} \iff 4 \geq 3 + 2\mu_4 \iff \mu_4 \leq \frac{1}{2},$$

$$A \text{ is optimal} \iff 3 + 2\mu_4 \geq 4 \iff \mu_4 \geq \frac{1}{2}.$$

Suppose Player 2 plays pure $D$ at $n = 4$. Then at $n = 3$ the rational type of Player 1 compares

$$D \mapsto 2, \qquad A \mapsto 1,$$

so he strictly prefers $D$. Thus only the crazy type reaches $n = 4$, implying

$$\mu_4 = 1.$$

But pure $D$ at $n = 4$ requires $\mu_4 \leq \frac{1}{2}$, a contradiction.
Suppose instead that Player 2 plays pure $A$ at $n = 4$. Then at $n = 3$ the rational type of Player 1 compares

$$D \mapsto 2, \qquad A \mapsto 3,$$

so he strictly prefers $A$. Hence both types of Player 1 play $A$ at $n = 3$, and Bayes' rule implies

$$\mu_4 = \mu_2.$$

Since pure $A$ at $n = 4$ requires $\mu_4 \geq \frac{1}{2}$, it follows that

$$\mu_2 \geq \frac{1}{2}.$$

Then Player 2 also chooses $A$ at $n = 2$, so the rational type of Player 1 chooses $A$ at $n = 1$, implying

$$\mu_2 = \mu_0 = \frac{1}{3},$$

a contradiction.
Therefore Player 2 must mix at $n = 4$, and so

$$\mu_4 = \frac{1}{2}.$$

**Step 3: Verify the rational type's mixing probability at $n = 3$.**
Let $p_A$ be the probability with which the rational type of Player 1 chooses $A$ at $n = 3$. The crazy type chooses $A$ with probability 1.
Since both types choose $A$ at $n = 1$, Bayes' rule gives

$$\mu_2 = \mu_0 = \frac{1}{3}.$$

At $n = 4$, Bayes' rule yields

$$\mu_4 = \frac{\mu_2}{\mu_2 + (1 - \mu_2)p_A}.$$

Substituting $\mu_2 = \frac{1}{3}$ and $\mu_4 = \frac{1}{2}$,

$$\frac{1}{2} = \frac{\frac{1}{3}}{\frac{1}{3} + \frac{2}{3}p_A}.$$

Solving,

$$p_A = \frac{1}{2}.$$

**Step 4: Verify Player 2's mixing probability at $n = 4$.**
Let $q_A$ be the probability with which Player 2 chooses $A$ at $n = 4$.
At $n = 3$, if the rational type of Player 1 chooses $D$, he gets

$$2.$$

If he chooses $A$, then: - with probability $q_A$, Player 2 continues and he then chooses $D$ at $n = 5$, giving payoff 3; - with probability $1 - q_A$, Player 2 stops at $n = 4$, giving payoff 1.
So the expected payoff from $A$ is

$$q_A \cdot 3 + (1 - q_A) \cdot 1 = 1 + 2q_A.$$

Indifference requires

$$1 + 2q_A = 2,$$

so

$$q_A = \frac{1}{2}.$$

**Step 5: Verify that Player 2 prefers $A$ at $n = 2$.**
If Player 2 chooses $D$ at $n = 2$, she gets

$$3.$$

If she chooses $A$, then with probability $\frac{1}{3}$ Player 1 is crazy and with probability $\frac{2}{3}$ Player 1 is rational. If Player 1 is rational, continuation yields

$$\frac{1}{2} \cdot 2 + \frac{1}{2}\left(\frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 3\right) = \frac{11}{4}.$$

If Player 1 is crazy, continuation yields

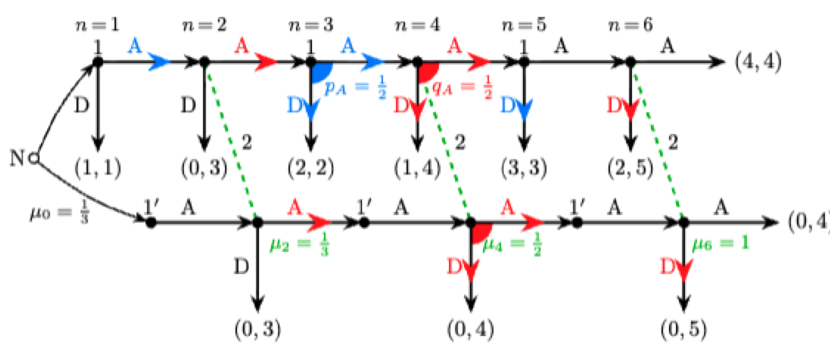
$$\frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 5 = \frac{9}{2}.$$

Hence the expected payoff from $A$ at $n = 2$ is

$$\frac{2}{3} \cdot \frac{11}{4} + \frac{1}{3} \cdot \frac{9}{2} = \frac{10}{3} > 3.$$

Therefore Player 2 strictly prefers $A$ at $n = 2$.

**Step 6: Verify that the rational type prefers $A$ at $n = 1$.**




Since Player 2 chooses $A$ at $n = 2$, the rational type of Player 1 prefers continuing at $n = 1$ to stopping immediately. Thus he chooses $A$ at $n = 1$, and this is consistent with

$$\mu_2 = \mu_0 = \frac{1}{3}.$$

**Conclusion.**
The following assessment is a PBE:

$$\mu_2 = \frac{1}{3}, \qquad \mu_4 = \frac{1}{2}, \qquad \mu_6 = 1,$$

$$\text{rational Player 1: } A \text{ at } n = 1, \text{ mix at } n = 3 \text{ with probability } \frac{1}{2} \text{ on } A, \; D \text{ at } n = 5,$$

$$\text{Player 2: } A \text{ at } n = 2, \text{ mix at } n = 4 \text{ with probability } \frac{1}{2} \text{ on } A, \; D \text{ at } n = 6.$$

The intuition is that the rational type of Player 1 initially mimics the crazy type in order to sustain doubt. That doubt induces Player 2 to continue early in the game. Near the end, however, the incentives unravel and both players stop.

**Example 4.23 (Von Neumann's model of bluffing in Poker).** Ann and Bob each privately draws a card; the value of Ann's card is $X$, the value of Bob's card is $Y$. Assume that $X$ and $Y$ are independent draws from the uniform distribution on $[0, 1]$. So, for example, $\Pr[a \leq X \leq b] = b - a$, and $\mathbb{E}[X \mid a \leq X \leq b] = (a + b)/2$, for any $0 \leq a \leq b \leq 1$, where "Pr" stands for probability, and "$\mathbb{E}$" for expected value.
The following are the rules of a rather simplified poker game.
Each player has to put £1 into the pot. Then Ann, knowing the realization of $X$ but not that of $Y$, either *checks* or *raises*. If she checks, then whoever has the highest card wins the pot (£1+£1=£2), and the game ends. Ignore $X = Y$ throughout as it occurs with zero probability.
If Ann raises, then she has to put another £1 into the pot, and it is Bob's turn: he either *folds* or *calls*.
If Bob folds, then Ann wins the pot (£2 put in by her, £1 by Bob, £3 in total) irrespective of the values of $X$ and $Y$, and the game ends. If Bob calls, then he has to put an additional £1 into the pot as well; their cards are compared, whoever has the highest card wins the pot (£2+£2=£4), and the game ends.
Both are aware of the structure of the game and are risk neutral.

(a) Explain (in a sentence or two) why, in any PBE, Bob's strategy is a threshold strategy, that is, there exists $c \in [0, 1]$ such that he calls if, and only if, $Y \geq c$. From Ann's perspective, what is the probability that Bob *folds* using threshold $c$?
If Ann checks, expected payoff is $\mathbb{P}(Y < X|X) - \mathbb{P}(X < Y|X) = X - (1 - X) = 2X - 1$. Bob's expected payoff is $\mathbb{P}(X < Y|Y) - \mathbb{P}(Y < X|Y) = 2Y - 1$.
If Ann raises and Bob calls, then Ann's expected payoff is

$$2\mathbb{P}(Y < X|X) - 2\mathbb{P}(X < Y|X) = 2X - 2(1 - X) = 4X - 2$$

while Bob's expected payoff is

$$2\mathbb{P}(X < Y|Y, \text{raise}) - 2\mathbb{P}(Y < X|Y, \text{raise}) = 2Y - 2(1 - Y) = 4Y - 2$$

where we can disregard raise since it is a function of $X$ only and hence independent of $Y$. Now we can answer the question.
Suppose Ann raises and Bob observes $Y = c$. Bob prefers call if

$$4\mathbb{P}(X < Y|Y = c, \text{raise}) - 2 \geq -1 \implies \mathbb{P}(X < Y|Y = c, \text{raise}) \geq \frac{1}{4}$$

Since $X$ is independent of $Y$

$$\mathbb{P}(X < c|\text{raise}) \geq \frac{1}{4}$$

which is increasing in $c$ by properties of the cdf. Hence, there is some threshold $c$ such that when $Y \geq c$, Bob prefers to call for $c \in [0, 1]$.
From Ann's perspective, probability Bob folds is

$$\mathbb{P}(Y < c) = c$$

since $Y \sim U[0, 1]$

(b) Suppose that Ann's draw is as bad as can be: $X = 0$. Verify that she prefers to raise provided $c > \frac{1}{3}$. Interpret.
If $X = 0$, then $u_A(\text{ch}|X = 0) = -1$ since $Y > 0$ w.p. 1. Ann's payoff when raising instead is the expected value (Bob folds and she gains 1 + Bob calls and she loses 2)

$$\begin{aligned} u_A(r|X = 0) &= \mathbb{P}(Y < c) + (1 - \mathbb{P}(Y < c))(-2) \\ &= c + (1 - c)(-2) \\ &= 3c - 2 \end{aligned}$$

Raising is better iff $3c - 2 > -1 \implies c > \frac{1}{3}$. Hence, bluffing with a terrible hand is profitabler if Bob's call threshold is sufficiently high i.e. folding probability is high.

(c) Derive a PBE in which Ann raises with $X \in [0, a]$ as well as $X \in [b, 1]$ but checks with $X$ between $a$ and $b$; conditional on Ann raising, Bob folds with $Y < c$ and calls with $Y > c$, where $0 < a < c < b < 1$ and $c > \frac{1}{3}$.
Hints: Write down three equations: Ann prefers to raise with $X \leq a$, she prefers to raise with $X \geq b$, and Bob is indifferent between folding and calling with $Y = c$. When writing down the last condition, keep in mind that it is conditional on the event that Ann has raised, i.e. $X \leq a$ or $X \geq b$. For instance, the probability that $X \leq a$ conditional on this event is

$$\Pr[X \leq a \mid X \leq a \text{ or } X \geq b] = \frac{a}{a+1-b}.$$

Compute the equilibrium values of $a < c < b$, verify that $c > \frac{1}{3}$.
If Ann checks with $x$ wins iff $x > Y$ which happens with probability $x$. Hence, $u_A(x, \text{ch}) = x \cdot 1 + (1 - x)(-1) = 2x - 1$. If Ann raises and Bob uses threshold $c$,

- with probability $c$, Bob folds
- with probability $1 - c$, Bob calls and $Y$ is uniform on $[c, 1]$

If $x \leq c$, conditional on call, Ann cannot win so

$$u_A(x, r) = c + (1 - c)(-2) = 3c - 2$$

If $x > c$, conditional on call,

$$\mathbb{P}(\text{Ann wins}|\text{call}) = \mathbb{P}(Y < x|Y \geq c) = \frac{x-c}{1-c}$$

Therefore, $u_A(x, r) = c + (1 - c)\left(4 \cdot \frac{x-c}{1-c} - 2\right) = c + 4(x - c) - 2(1 - c) = 4x - 2 - c$. Hence

$$u_A(x, r) = \begin{cases} 3c - 2, & x \leq c \\ 4x - 2 - c, & x > c \end{cases}$$

Since Ann raises below $a$ but checks just above $a$, indifferent at $x = a$.
Since $a < c$, $u_A(a, r) = 3c - 2 = u_A(\text{ch}, a) = 2a - 1$. Hence $a = \frac{3c-1}{2}$ and so $a^*(c)$ is the best reply to $c$.
Similarly for $x = b$, $u_A(b, r) = 4b - 2 - c = u_A(b, \text{ch}) = 2b - 1$ which implies $b = \frac{1+c}{2}$.
Bob is indifferent at $y = c$. From (a), $\mathbb{P}(X < Y|\text{raise}) = \frac{1}{4}$ at indifference. At $y = c$, $\mathbb{P}(X < c|\text{raise}) = \frac{1}{4}$.
Since $a < c$ and raise iff $X \in [0, a] \cup [b, 1]$,

$$\mathbb{P}(X < c|\text{raise}) = \mathbb{P}(X < a|X < a \text{ or } X > b) = \frac{a}{a + (1 - b)}$$

Hence

$$\frac{1}{4} = \frac{a}{a+1-b} \implies 4a = a + 1 - b \implies b = 1 - 3a$$

Solving, we have

$$\begin{aligned} \frac{1+c}{2} &= 1 - 3 \cdot \frac{3c-1}{2} \\ 1 + c &= 2 - 9c + 3 \\ 10c &= 4 \end{aligned}$$

which yields

$$a = \frac{1}{10}, b = \frac{7}{10}, c = \frac{2}{5}$$

when $c = \frac{2}{5} > \frac{1}{3}$. So the PBE is

- Ann raises iff $X \in [0, \frac{1}{10}] \cup [\frac{7}{10}, 1]$, Bob calls iff $Y \geq c = \frac{2}{5}$.
- Bob's belief after raising is $\mu_B(X \in [0, \frac{1}{10}]|r) = \frac{1}{4}$ and $\mu_B(X \in [\frac{7}{10}, 1]|r) = \frac{3}{4}$ while Ann's beliefs are common prior $Y \sim U[0, 1]$

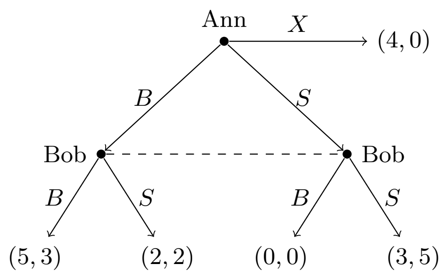
(d) Would you rather be Ann or Bob in this poker game?
Rather be Ann since ex-ante payoffs are positive while Bob's ex ante payoffs are negative and you can bluff to use dishonest communication. Bob has to decide under uncertainty.

# 5 Refinements of Dynamic Games

## 5.1 Forward Induction, Intuitive Criterion

**Motivation**

**Example 5.1.** Consider BoS game where Ann has an attractive option.




| | $B$ | $S$ |
|---|---|---|
| $B$ | $5,3$ | $2,2$ |
| $S$ | $0,0$ | $3,5$ |
| $X$ | $4,0$ | $4,0$ |

Intuitively, Ann's outside option, with a payoff that beats coordinating on $S$ should help her achieve $(B,B)$ because Bob must know that Ann would not come into the game unless she expects to get more than 4. But $(X,S)$ is Nash/SPE and also PBE.

**How $(X,S)$ can be sustained as a PBE**

If Bob believes $\mu_S \geq 1/6$, then he plays $S$ since his payoff would be $5\mu_S + (1-\mu_S)2 = 3\mu_S + 2 \geq 3(1-\mu_S)$.
Anticipating that, Ann plays $X$ since she yields less than 4.
Note this is not the only PBE since $(B,B)$ with $\mu_S = 0$ is also a PBE.

**How $(X,S)$ can be sustained as a Sequential Equilibrium**

For $m = 1, 2, \dots$, let $\sigma_A^m = \frac{1}{m^2}B + \frac{1}{m}S + (1 - \frac{1}{m^2} - \frac{1}{m})X$ and let $\sigma_B^m = \frac{1}{m}B + (1-\frac{1}{m})S$. Using Bayes's rule, Bob's belief that Ann has played $S$ conditional on Ann having played either $B$ or $S$ by mistake is

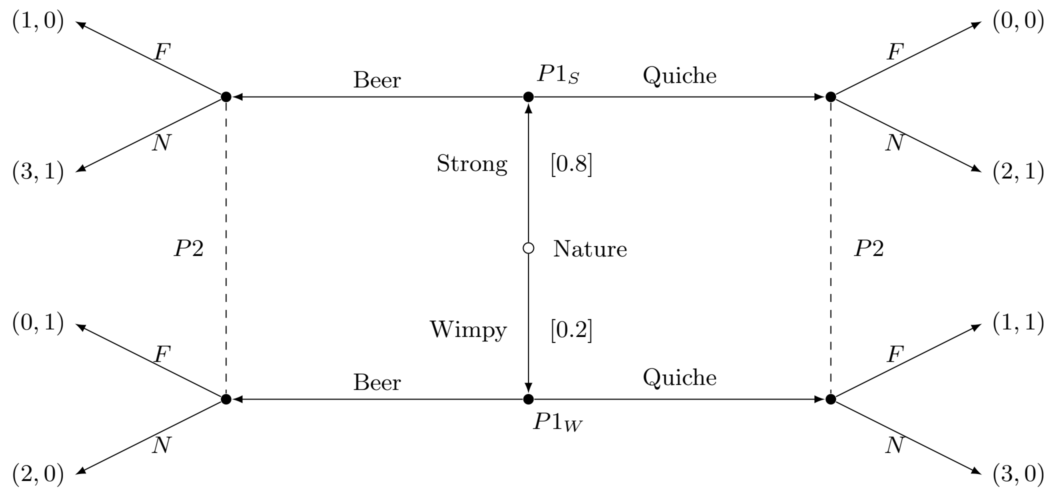
$$\mu_S^m = \mathbb{P}[S|\text{not X}] = \frac{1/m}{1/m^2 + 1/m} = \frac{m}{m+1}$$

As $m \to \infty$: $\sigma_A^m \to X, \sigma_B^m \to S, \mu_S^m \to 1$ hence $(X,S)$ with $\mu_S = 1$ is sequential equilibrium.

**Why this is counterintuitive**: But Ann guarantees 4 by playing $X$ so by playing $X$ she must be expecting more. Only $B$ (followed by $B$) can yield more for her so intuitively, we should let $\mu_S = 0$. This is a forward induction argument

**Definition 5.2 (Intuitive Criterion).** Upon observing a deviation, we must not put positive probability (beliefs) on types whose equilibrium payoff exceeds all the possible payoffs that they could get from the deviation, assuming the opponent's reaction to the deviation is rational on some beliefs

**Example 5.3 (Beer-Quiche game). TikZ extensive-form tree**




**Interpretation of the tree**

Nature first chooses Player 1's type: strong ($S$) with probability 0.8 and wimpy ($W$) with probability 0.2. Player 1 observes his type and chooses Beer ($B$) or Quiche ($Q$). Player 2 observes only the breakfast choice, not the type, so the two nodes following $B$ form one information set and the two nodes following $Q$ form another information set. After observing the breakfast choice, Player 2 chooses Fight ($F$) or Not fight ($N$).

**Normal-form representation**

A pure strategy for Player 1 specifies what each type does:

$$\Sigma_1 = \{BB, BQ, QB, QQ\},$$

where the first letter is the strong type's action and the second letter is the wimpy type's action. Thus:

$$BB : (S \to B,\ W \to B), \quad BQ : (S \to B,\ W \to Q), \quad QB : (S \to Q,\ W \to B), \quad QQ : (S \to Q,\ W \to Q).$$

A pure strategy for Player 2 specifies what to do after each observed breakfast:

$$\Sigma_2 = \{FF, FN, NF, NN\},$$

where the first letter is the action after $B$ and the second letter is the action after $Q$. Thus:

$$FF : (B \to F,\ Q \to F), \quad FN : (B \to F,\ Q \to N), \quad NF : (B \to N,\ Q \to F), \quad NN : (B \to N,\ Q \to N).$$

Expected payoffs are computed using the prior probabilities 0.8 and 0.2.

| | $FF$ | $FN$ | $NF$ | $NN$ |
|---|---|---|---|---|
| $BB$ | $(0.8, 0.2)$ | $(1.2, 0.0)$ | $(2.4, 1.0)$ | $(2.8, 0.8)$ |
| $BQ$ | $(1.0, 0.2)$ | $(1.4, 0.0)$ | $(2.6, 1.0)$ | $(3.0, 0.8)$ |
| $QB$ | $(0.0, 0.2)$ | $(0.4, 0.0)$ | $(1.6, 1.0)$ | $(2.0, 0.8)$ |
| $QQ$ | $(0.2, 0.2)$ | $(0.6, 0.0)$ | $(1.8, 1.0)$ | $(2.2, 0.8)$ |

**How these normal-form payoffs are computed**

For example, consider $(BQ, FN)$.

Under $BQ$, the strong type chooses $B$ and the wimpy type chooses $Q$. Under $FN$, Player 2 fights after $B$ and does not fight after $Q$. Therefore:

$$u_1(S, B, F) = 1, \qquad u_1(W, Q, N) = 3,$$

$$u_2(S,B,F)=0, \qquad u_2(W,Q,N)=0.$$

Taking expectations using $\Pr(S)=0.8$ and $\Pr(W)=0.2$ gives

$$U_1(BQ,FN)=0.8(1)+0.2(3)=1.4,$$

$$U_2(BQ,FN)=0.8(0)+0.2(0)=0.$$

**PBE analysis**

Let

$$\mu(a)=\Pr(S\mid a)$$

be Player 2's posterior belief after observing breakfast choice $a\in\{B,Q\}$.

If Player 2 observes action $a$, then:

$$\mathbb{E}[u_2(F\mid a)]=1-\mu(a), \qquad \mathbb{E}[u_2(N\mid a)]=\mu(a).$$

Hence:

$$F \text{ is optimal } \iff \mu(a)\le \tfrac{1}{2}, \qquad N \text{ is optimal } \iff \mu(a)\ge \tfrac{1}{2}.$$

**No separating PBE**

First check $S\to B$, $W\to Q$, and so in a separating equilibrium, Player 1's choice reveals information about his type. Then Bayes' rule implies

$$\mu(B)=1, \qquad \mu(Q)=0.$$

So Player 2 chooses $N$ after $B$ and $F$ after $Q$. But then the wimpy type gets

$$u_1(W,Q,F)=1$$

from staying with $Q$, and

$$u_1(W,B,N)=2$$

from deviating to $B$. Thus the wimpy type deviates.

Next check $S\to Q$, $W\to B$. Then Bayes' rule implies

$$\mu(Q)=1, \qquad \mu(B)=0.$$

So Player 2 chooses $N$ after $Q$ and $F$ after $B$. But then the wimpy type gets

$$u_1(W,B,F)=0$$

from staying with $B$, and

$$u_1(W,Q,N)=3$$

from deviating to $Q$. Thus the wimpy type deviates again.

Therefore there is no separating PBE.

**Pooling on Beer**

Suppose both types choose Beer and so the posterior does not change on-path:

$$S\to B, \qquad W\to B.$$

Then Bayes' rule implies

$$\mu(B)=0.8.$$

Since $0.8 > \frac{1}{2}$, Player 2 chooses

$$N \text{ after } B.$$

Quiche is off path, so Bayes' rule does not pin down $\mu(Q)$. Choose off-path beliefs with

$$\mu(Q) \leq \tfrac{1}{2},$$

so that Player 2 chooses

$$F \text{ after } Q.$$

This is because the expected payoff

$$\mathbb{E}[u_2(F|Q)] = 0 \cdot \mu(Q) + (1 - \mu(Q)) = 1 - \mu(Q)$$

and

$$\mathbb{E}[u_2(N|Q)] = \mu(Q) + 0 \cdot (1 - \mu(Q))$$

and so, P2 chooses $F$ just when

$$\mu(Q) \leq \frac{1}{2}$$

Check incentives:

$$u_1(S, B, N) = 3 > 0 = u_1(S, Q, F),$$
$$u_1(W, B, N) = 2 > 1 = u_1(W, Q, F).$$

So neither type deviates.

Hence pooling on Beer is a PBE.

**Pooling on Quiche**

Suppose both types choose Quiche:

$$S \to Q, \qquad W \to Q.$$

Then Bayes' rule implies

$$\mu(Q) = 0.8.$$

Since $0.8 > \frac{1}{2}$, Player 2 chooses

$$N \text{ after } Q.$$

Beer is off path, so choose beliefs with

$$\mu(B) \leq \tfrac{1}{2},$$

so that Player 2 chooses

$$F \text{ after } B.$$

since

$$\mathbb{E}[u_2(F|B)] = 0 \cdot \mu(B) + (1 - \mu(B))$$
$$\mathbb{E}[u_2(N|B)] = \mu(B) + 0 \cdot (1 - \mu(B))$$

and so to prevent incentives to deviate, to make Player 2 choose $F$ after $B$, we enforce

$$\mathbb{E}[u_2(F|B)] \geq \mathbb{E}[u_2(N|B)] \iff \mu(B) \leq \frac{1}{2}$$

Check incentives:

$$u_1(S, Q, N) = 2 > 1 = u_1(S, B, F),$$
$$u_1(W, Q, N) = 3 > 0 = u_1(W, B, F).$$

So neither type deviates.

Hence pooling on Quiche is also a PBE.

**Apply the Intuitive Criterion**

Start from the pooling-on-Quiche equilibrium. Equilibrium payoffs are

$$u_1(S, Q, N) = 2, \qquad u_1(W, Q, N) = 3.$$

Consider the off-path deviation to Beer.

For the wimpy type, the best possible payoff from deviating to Beer is

$$u_1(W, B, N) = 2,$$

which is still less than the equilibrium payoff 3. Therefore the wimpy type can never benefit from deviating to Beer.

By the Intuitive Criterion, after observing Beer off path, Player 2 should place zero probability on the wimpy type. Hence:

$$\mu(B) = 1.$$

Then Player 2's best response after Beer is

$$N.$$

But now the strong type gets

$$u_1(S, B, N) = 3 > 2 = u_1(S, Q, N),$$

so the strong type strictly prefers to deviate.

Therefore pooling on Quiche is eliminated by the Intuitive Criterion.

**Why pooling on Beer survives**

Under pooling on Beer, equilibrium payoffs are

$$u_1(S, B, N) = 3, \qquad u_1(W, B, N) = 2.$$

Consider the off-path deviation to Quiche.

The strong type can get at most

$$u_1(S, Q, N) = 2 < 3,$$

so the strong type can never benefit from deviating to Quiche.

The wimpy type, however, could get

$$u_1(W, Q, N) = 3 > 2.$$

So only the wimpy type could plausibly benefit from deviating to Quiche. The Intuitive Criterion therefore allows Player 2 to infer

$$\mu(Q) = 0.$$

Then Player 2 fights after Quiche, making deviation unattractive. Hence pooling on Beer survives (also just note that $\mu(Q) = 1 \geq \frac{1}{2}$ so it naturally satisfies the off-path belief restriction for the PBE of pooling on beer).

**Conclusion**

The game has two pooling PBEs and no separating PBE. The Intuitive Criterion eliminates pooling on Quiche and selects the unique refined outcome:

$$\boxed{\text{Both types choose Beer, and Player 2 does not fight after Beer.}}$$

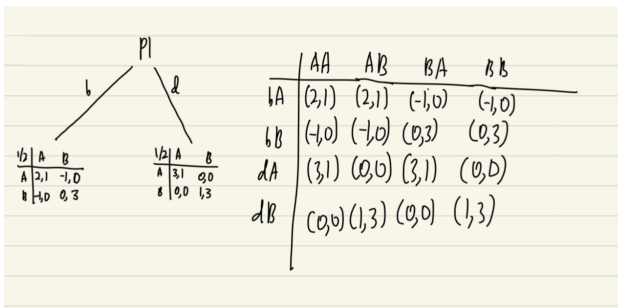
**Example 5.4 (Burning).** Consider a coordination game where P1 and P2 simultaneously choose $A$ or $B$. If both choose $A$ then P1 gets 3 utils, P2 gets 1; if both choose $B$ then P1 gets 1 util, P2 gets 3. They both get zero utility in all other cases.

(a) Write down this game and find all Nash equilibria.

$$\begin{array}{c|cc} & A & B \\ A & 3,1 & 0,0 \\ B & 0,0 & 1,3 \end{array}$$

Pure NE is $(A, A)$ and $(B, B)$. Mixed NE is $(\frac{3}{4}A + \frac{1}{4}B, \frac{1}{4}A + \frac{3}{4}B)$

(b) Now assume that before the game starts, it is known to both players that P1 can publicly "burn" 1 unit of his own utility; P2 observes this. (To clarify: if P1 burns a unit of utility then instead of 0 she starts with $-1$, etc.) Draw an extensive form, where you may indicate simultaneous-move subgames by an appropriate matrix. Write down the corresponding reduced-form strategic game (hint: it's $4 \times 4$).




(c) Find all pure-strategy Nash equilibria and SPE in the game considered in part (b).
By inspection:

- Pure NE: $\{(dA, AA), (bA, AB), (dB, BB), (dA, BA)\}$
- Pure SPNE: $\{(d; AA, AA), (b; AB, AB), (d; BA, BA), (d; BB, BB)\}$

(d) Show that only one subgame-perfect equilibrium survives the iterated maximal elimination of weakly dominated strategies. (That is, in every round, eliminate as many weakly dominated strategies as possible from both players.)
Claim: $(d; AA, AA)$ is the only SPNE that survives. We have the following procedure $dA \succ dB, AA \succ BA, AB \succ BB, bA \succ dB, AA \succ AB, dA \succ bA$

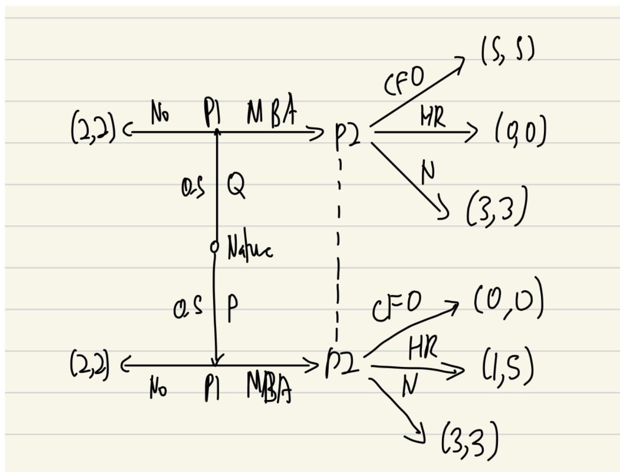
(e) Give a forward-induction argument to support the play of this equilibrium. Do you think that this argument is reasonable?
If P1 plays $d$ and expects $(B, B)$ P1's payoff is 1. If P1 plays $b$ and expects $(A, A)$ after $b$, P1's payoff is 2. So since P1 does not burn, he expects to get a payoff of 2, since otherwise he would play $b$ to coordinate on $A$.
So P2 knows that playing $d$ implies that P1 is coordinating on A
Reasonable if players interpret forgone costly actions as credible information about intentions.

**Example 5.5 (MBA or no MBA).** P1 (he) is a recent graduate of the University of Oxford employed in the City. He has a privately known type, either *Quant* (type $Q$) or *Poet* (type $P$), with 50-50% prior probabilities. He has the opportunity to pursue an MBA degree in Finance back at Oxford. If he does this then his employer (P2) can promote him to Chief Financial Officer (CFO), which is a good match for the $Q$ type, or Head of Human Resources (HR), which is the best use of a $P$ type, or just keep him in his current position with a raise. Neither type of P1 likes to be promoted to the HR job because it involves firing people. If P1 does not get his MBA degree then he cannot be promoted.

The payoffs are summarized in the table below. Thinking of it as the matrix representation of a normal-form game, you may interpret it as follows. Nature chooses the row (P1's type, $Q$ or $P$, with 50-50%), P1 chooses the matrix (No or MBA), and P2 chooses the column (conditional on MBA, either CFO, or HR, or N). Payoffs are listed for P1 first, P2 second.

| | No | MBA | | |
|---|---|---|---|---|
| | | CFO | HR | N |
| [50%] $Q$ | $(2,2)$ | $(5,5)$ | $(0,0)$ | $(3,3)$ |
| [50%] $P$ | $(2,2)$ | $(0,0)$ | $(1,5)$ | $(3,3)$ |

(a) Draw the corresponding extensive form (game tree), including the relevant (non-singleton) information sets.




(b) Derive all pure-strategy perfect Bayesian equilibria of the game.
Define P2's belief $\mu := \mathbb{P}(Q|\text{MBA})$.
P2's payoffs are

- $u_2(\text{CFO}) = 5\mu$
- $u_2(\text{HR}) = 5(1-\mu)$
- $u_2(N) = 3$

so,

$$BR_2(\mu) = \begin{cases} \text{CFO} & \text{if } \mu > \frac{3}{5} \\ \text{HR} & \text{if } \mu < \frac{2}{5} \\ N & \text{if } \frac{2}{5} \le \mu \le \frac{3}{5} \end{cases}$$

For P1, Type Q prefers MBA if P2 plays CFO ($5 > 2$) or N ($3 > 2$) and prefers No if P2 plays HR ($0 < 2$). Type P prefers MBA if P2 plays N ($3 > 2$); prefers No if P2 plays CFO ($0 < 2$) or HR ($1 < 2$). We have three PBE

- Pooling on MBA where $\mu = \mathbb{P}(Q|\text{MBA}) = \mathbb{P}(Q) = \frac{1}{2}$. Notice that by Bayes Rule

$$\mu(Q|\text{MBA}) = 0.5$$

  So, P2's best response is to play $N$ and P1 receives 3. Neither type has incentive to deviate since deviating to No yields 2.
- Pooling on No. Notice that Bayes Rule does not pin down $\mu(Q|\text{MBA})$ since MBA is off-path. But to prevent deviation, we want P2 to play HR. So, we need

$$\mu(Q|\text{MBA}) < \frac{2}{5}$$

- Separating with $Q \to$ MBA, $P \to$ No. Hence

$$\mu = \mathbb{P}(Q|\text{MBA}) = 1$$

P2's best response is CFO. For type Q: MBA gives $5 > 2$ so Q chooses MBA.
Type P: if he deviates to MBA, P2 still chooses CFO since $\mu = 1$. Since $0 < 2$, type P has no profitable deviation. So PBE is $(Q:\ \text{MBA}, P:\ \text{No};\ \text{CFO})$ with $\mu = 1$

(c) Define the Intuitive Criterion (equilibrium dominance). Does it eliminate the equilibrium where both types of P1 play No? Why or why not? In pooling on No, each type gets equilibrium payoff 2. If type P deviates to MBA,

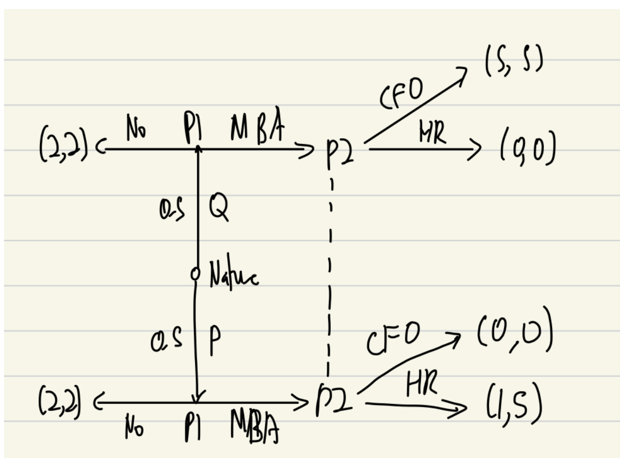
- If P2 plays HR, P gets $1 < 2$
- If P2 plays CFO, P gets $0 < 2$
- If P2 plays N, P gets $3 > 2$

So deviation is not equilibrium dominated. If type Q deviates

- If P2 plays HR, Q gets $0 < 2$
- If P2 plays CFO, Q gets $5 > 2$
- If P2 plays N, Q gets 3

This deviation is not equilibrium dominated either. N is sequentially rational reply for $\mu \in [0.4, 0.6]$ so P can benefit from deviating to MBA under a rational P2 reply (for that belief).
Hence IC does not eliminate pooling on No

(d) Suppose P2 does not have action N (cannot keep the employee in his current position, even with a raise, in case he gets an MBA). Does the Intuitive Criterion eliminate pooling on No?




Suppose P1 pools on No, receives payoff 2 for both types. If type P deviates

- CFO: $0 < 2$
- HR: $1 < 2$

so deviation to MBA for P is equilibrium dominated. Hence, we may assign $\mu = \mathbb{P}(Q|\text{MBA}) = 1$ as $1 - \mu = \mathbb{P}(P|\text{MBA}) = 0$ because type $P$ would never deviate to MBA ever for any belief that P2 has. Then $(Q:\ \text{No}, P:\ \text{No};\ \text{CFO})$ with $\mu = 1$ since

$$BR_2 = \begin{cases} \text{CFO} & \text{if } \mu \geq \frac{1}{2} \\ \text{HR} & \text{if } \mu < \frac{1}{2} \end{cases}$$

Then pooling on No is not a PBE since Q deviating to MBA would guarantee P2 playing CFO since $\mu = 1$. Hence pooling on No is eliminated by IC and separating equilibrium is the only PBE that remains.

## 5.2 Trembling Hand Perfect Equilibrium

The main motivation for this set of refinements is about what to infer from a deviation. Under one body of literature, players interpret deviations as accidental mistakes.

**Definition 5.6 (Trembling-hand perfect equilibrium in strategic games).** A trembling hand perfect equilibrium of a finite strategic game is a mixed strategy profile $\sigma$ with the property that there exists a sequence $(\sigma^k)_{k=}^{\infty}$ of completely mixed strategy profiles that converge to $\sigma$ such that for each player $i$ the strategy $\sigma_i$ is a best response to $\sigma_{-i}^k$ for all values of $k$

**Proposition 5.7.** A strategy profile in a finite two player strategy game is a trembling hand equilibrium iff it is a mixed strategy NE and the strategy of neither player is weakly dominated. Furthermore, every finite strategic game has a trembling hand perfect equilibrium

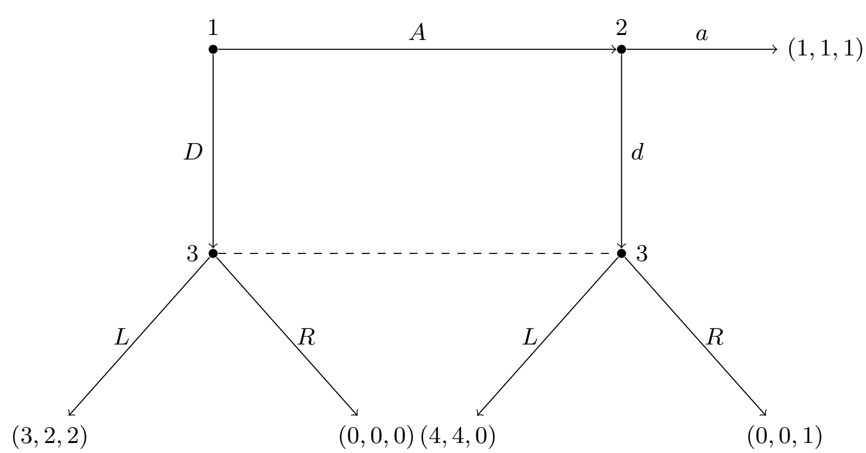
**Definition 5.8.** A trembling hand perfect equilibrium of a finite extensive game is a behavioural strategy profile that corresponds to a trembling hand perfect equilibrium of the reduced strategic form of the game.

**Remark.** A behavioural strategy (Osbourne and Rubinstein) is a collection $(\sigma_i(I_i))_{i\in\mathcal{I}_i}$ of independent probability measures where $\sigma_i(I_i)$ is a probability measure over $A(I_i)$. In other words, a behavioural strategy specifies a probability measure over the actions available to player $i$ at each of his information sets.

**Proposition 5.9.** For every trembling hand perfect equilibrium $\sigma$ of a finite extensive game with perfect recall, there is a belief system $\mu$ such that $(\sigma, \mu)$ is a sequential equilibrium of the game

**Example 5.10 (Selten's Horse Revisited).** Selten's Horse is a three-player extensive form game that illustrates why **subgame perfect equilibrium (SPE)** is insufficient to rule out unreasonable equilibria, and how **trembling hand perfect equilibrium (THPE)** and **perfect Bayesian equilibrium (PBE)** provide sharper refinements.

**The Game Tree**




**Description.** Player 1 moves first: she can either *stop* by playing $A$ (yielding payoffs $(1, 1, 1)$) or *continue* by playing $D$, passing the move to Player 2. Player 2 can similarly stop with $A$ (yielding $(0, 0, 0)$) or continue with $D$, reaching Player 3's information set. Player 3 then chooses $L$ or $R$. Crucially, **Player 3 cannot distinguish** whether she was reached because Player 2 played $D$ on the equilibrium path, or due to some off-path deviation — hence the dashed information set.

**Step 1: An Unreasonable SPE.**
Consider the strategy profile:
$$\sigma^* = \big(A,\ D,\ R\big)$$
That is, Player 1 plays $A$, Player 2 plays $D$, and Player 3 plays $R$.

- Player 1 gets 1 from $A$. If she deviated to $D$, play would reach P2, who plays $D$, then P3 who plays $R$, giving Player 1 a payoff of 0. So Player 1 has no incentive to deviate. ✓
- Player 2's strategy $D$ is never reached (Player 1 stops), so any strategy for P2 is trivially optimal. ✓
- Player 3's strategy $R$ is never tested on the equilibrium path. Since P3's information set is *off-path*, her belief is unrestricted by Bayes' rule, and $R$ is consistent with some belief. ✓

This is a valid SPE — yet it relies on Player 3's *threat* to play $R$, which is what deters Player 1 from playing $D$. But is this threat credible?

**Step 2: Why $R$ is Not Credible — Trembling Hand.**
The **trembling hand** argument asks: what if every player makes *small mistakes*? Formally, consider a sequence of fully mixed strategy profiles $\sigma^\varepsilon \to \sigma^*$ in which every action is played with at least probability $\varepsilon > 0$.
Under $\sigma^\varepsilon$, Player 3's information set is **reached with positive probability** (since P1 and P2 both tremble toward $D$ with probability $\varepsilon$). Player 3 must therefore best-respond. Comparing Player 3's payoffs:
$$u_3(L) = 2 \quad > \quad u_3(R) = 1$$
Player 3 strictly prefers $L$ regardless of her belief over nodes in her information set (both nodes yield the same payoffs). Hence, in *any* fully mixed approximation, Player 3 plays $L$. The strategy $R$ cannot survive trembling hand: $\sigma^*$ is **not trembling hand perfect**.

**Step 3: The Trembling Hand Perfect Equilibrium.**
The unique THPE is:
$$\sigma^{**} = \big(D,\ D,\ L\big)$$

- **Player 3** plays $L$ (strictly dominant at her info set, as shown above).
- **Player 2**, anticipating $L$ from Player 3, gets $u_2(D) = 2 > u_2(A) = 0$, so P2 plays $D$.
- **Player 1**, anticipating $D$ from P2 and $L$ from P3, gets $u_1(D) = 3 > u_1(A) = 1$, so P1 plays $D$.

Backward induction under the *trembling hand discipline* uniquely selects $(D, D, L)$ with payoffs $(\mathbf{3}, \mathbf{2}, \mathbf{2})$.

**Step 4: Consistency with PBE.**
A PBE requires (i) sequential rationality and (ii) beliefs updated by Bayes' rule wherever possible. In $\sigma^{**} = (D, D, L)$:

- Player 3's information set is **on the equilibrium path**, so Bayes' rule *does* pin down her belief: she assigns probability 1 to the upper node (P2 played $D$ as expected). Given this belief, $L$ is optimal. ✓
- The earlier unreasonable equilibrium $(A, D, R)$ fails PBE because $R$ is strictly dominated at P3's info set for *any* belief — a PBE requires sequential rationality, ruling it out. ✓

Thus, **THPE refines PBE**: every THPE is a PBE, but not vice versa. The trembling hand provides an independent *equilibrium selection* justification for why we should believe $(D, D, L)$ is the right prediction.

**Key Takeaway.**

| Solution Concept | Selects $(A, D, R)$? | Selects $(D, D, L)$? |
|---|---|---|
| Nash Equilibrium | ✓ | ✓ |
| Subgame Perfect Eq. | ✓ | ✓ |
| Perfect Bayesian Eq. | $\times$ | ✓ |
| Trembling Hand Perfect | $\times$ | ✓ |

The horse game demonstrates that off-path information sets require *discipline on beliefs and perturbations* — both PBE and THPE eliminate the unreasonable equilibrium, and the trembling hand provides an intuitive "mistakes" foundation for why $(D, D, L)$ is the uniquely credible outcome.

## 5.3 Signalling Games

**Motivation**: Employers are unable to observe types of employees, and thus employees need to signal their type by choosing certain actions (like getting a degree) in order to get higher pay. These high or low type employees either pool or separate on getting a degree.

**Example 5.11 (Spencian Signalling Game).** Spence's (1973) model: a privately informed sender (a worker) takes an action observable by an uninformed receiver (firms), who then updates beliefs and responds. The key insight is that even a *purely wasteful* action can serve as a credible signal of private information in equilibrium.

**Setup.**
The game has four stages:

1. **Nature** draws the worker's type $\theta \in \{L, H\}$, with $\Pr(\theta = H) = \lambda$, where $H > L \geq 0$.
2. The **worker** observes $\theta$ and chooses education level $e \geq 0$.
3. **Firms** (at least two, competing) observe $e$ but not $\theta$, and post wages $w$.
4. The worker accepts the best offer.

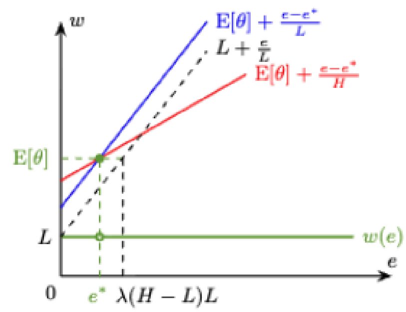
Payoffs: the worker earns $w - e/\theta$ (education is *less costly* for the high type); each firm earns $\theta - w$ if it hires the worker, and 0 otherwise. Competition between firms drives wages to $w(e) = \mathbb{E}[\theta \mid e]$.

**What is a PBE here?**
A PBE consists of strategies $\{e_L, e_H\}$, a wage function $w(e)$, and beliefs $\mu(e) = \Pr[\theta = H \mid e]$ for all $e \geq 0$, such that:

- Each type of worker maximises $w(e) - e/\theta$ given $w(\cdot)$.
- Firms set $w(e) = \mu(e) \cdot H + (1 - \mu(e)) \cdot L$.
- On-path beliefs are determined by Bayes's rule; off-path beliefs $\mu(e')$ are chosen freely by the modeller.

The freedom to set off-path beliefs is what generates a *multiplicity* of PBE, which the Intuitive Criterion will later discipline.

**Step 1: Pooling Equilibria.**
In a **pooling** equilibrium, both types choose the same education level $e^* = e_L = e_H$. Since both types pool, Bayes's rule pins down the on-path belief: $\mu(e^*) = \lambda$, so $w(e^*) = \lambda H + (1 - \lambda)L \equiv \mathbb{E}[\theta]$.
For any off-path $e' \neq e^*$, the modeller sets $\mu(e') = 0$, giving $w(e') = L$. This deters deviation: no worker wants to spend extra on education just to receive a lower wage.
The pooling action must satisfy $e^* \leq \lambda(H - L)L$, otherwise the low type would prefer to deviate to $e' = 0$ and receive $L$ rather than bear the cost of $e^*$.




**Step 2: Separating Equilibria.**
In a **separating** equilibrium, each type chooses a distinct education level, so firms can perfectly infer the worker's type from $e$. Bayes's rule therefore gives:

$$\mu(0) = 0 \implies w(0) = L, \qquad \mu(e_H) = 1 \implies w(e_H) = H.$$

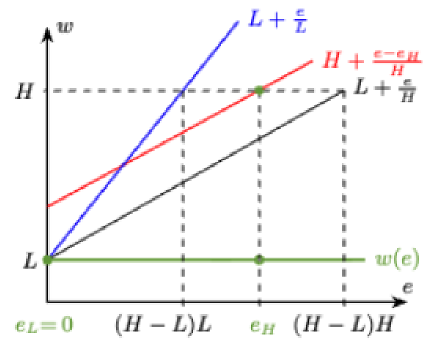
The low type has no incentive to signal, so $e_L = 0$ (spending on education only to be identified as low type is wasteful). The high type must choose $e_H$ large enough that the low type does not want to mimic:

$$\underbrace{H - \frac{e_H}{L}}_{\text{low type mimics}} \leq \underbrace{L}_{\text{low type stays at } e=0} \implies e_H \geq (H-L)L.$$

Symmetrically, the high type must prefer $e_H$ over $e = 0$:

$$H - \frac{e_H}{H} \geq L \implies e_H \leq (H-L)H.$$

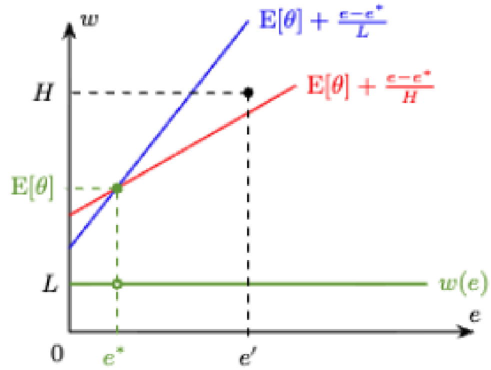
Any $e_H \in [(H-L)L,\ (H-L)H]$ supports a separating PBE, with off-path beliefs $\mu(e') = 0$ for all $e' \neq e_H$.



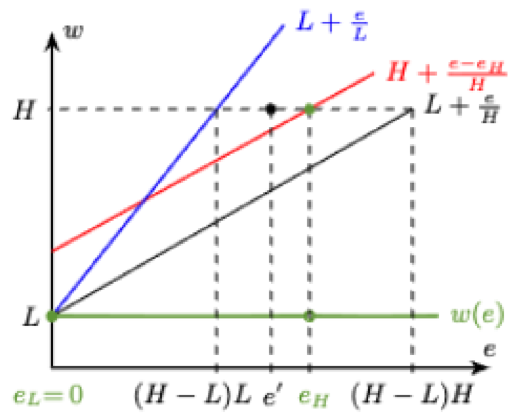

**Step 3: Applying the Intuitive Criterion.**
The multiplicity of equilibria above is unsatisfying. The **Intuitive Criterion** (Cho & Kreps, 1987) imposes discipline on off-path beliefs by asking: upon observing a deviation $e'$, could we rule out any type as its source?

> *Do not assign positive belief to type $\theta$ after observing deviation $e'$ if $\theta$'s equilibrium payoff strictly exceeds the* best *payoff $\theta$ could ever obtain from $e'$, given any rational response by the firm.*

**Ruling out pooling.** In any pooling equilibrium at $e^*$, consider a deviation to some $e' > e^*$. The low type cannot possibly gain from this deviation: their cost $e'/L$ is too high for the wage increase to compensate. But the high type *could* gain, if the firm responds with $w = H$. The Intuitive Criterion therefore requires $\mu(e') = 1$ after such a deviation, giving $w(e') = H$. But then the high type strictly prefers to deviate, breaking the pooling equilibrium.




**Ruling out non-minimal separating equilibria.** Consider a separating equilibrium with $e_H > (H-L)L$. For any $e' \in ((H-L)L,\ e_H)$, the low type cannot gain from deviating to $e'$ (their payoff from $e'$ at wage $H$ is still below their equilibrium payoff of $L$). But the high type could gain. The Intuitive Criterion again forces $\mu(e') = 1$, so $w(e') = H$, and the high type deviates. Only the *minimal-cost* separating equilibrium at $e_H = (H-L)L$ survives.

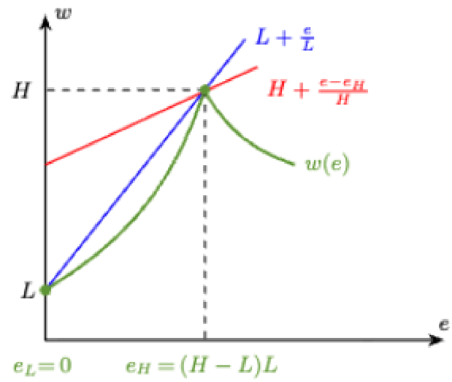



**Step 4: The Unique Surviving Outcome.**

The **minimal-cost separating equilibrium** (also called the **Riley outcome**) is the unique PBE surviving the Intuitive Criterion:

$$e_L = 0, \quad e_H = (H-L)L, \quad w(0) = L, \quad w(e_H) = H.$$

The high type distorts education *just enough* to make imitation unprofitable for the low type. No further separation is needed, and no pooling can survive.




**Key Takeaways.**

1. **PBE alone** yields a large multiplicity of pooling and separating equilibria, since off-path beliefs are unconstrained.
2. **The Intuitive Criterion** eliminates pooling equilibria and all but the minimal-cost separating equilibrium, by requiring that off-path beliefs do not assign probability to types who could not possibly gain from the deviation.
3. **Signalling need not be costly per se** — what matters is that the action is *relatively* cheaper for the high type (*single-crossing property*). The education level itself need not raise productivity to serve as a signal.
4. The Riley outcome is **constrained inefficient**: the high type over-invests in education relative to the first-best, but this distortion is necessary to sustain credible separation.